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McKinnon; David et al.

Wide area augmented reality location-based services

Abstract

Apparatus, methods and systems of providing AR content are disclosed. Embodiments of the inventive subject matter can obtain an initial map of an area, derive views of interest, obtain AR content objects associated with the views of interest, establish experience clusters and generate a tile map tessellated based on the experience clusters. A user device could be configured to obtain and instantiate at least some of the AR content objects based on at least one of a location and a recognition.

Inventors: McKinnon; David (San Francisco, CA), Wnuk; Kamil (Playa del Rey, CA), Sudol; Jeremi (New York, NY), Siddiqui; Matheen (Culver City, CA), Wiacek; John (Los Angeles, CA), Song; Bing (La Canada, CA), Witchey; Nicholas J. (Laguna Hills, CA)

Applicant: Nant Holdings IP, LLC (Culver City, CA)

Family ID: 52825798

Assignee: Nant Holdings IP, LLC (Culver City, CA)

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References Cited**U.S. PATENT DOCUMENTS**

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3050870	12/1961	Heilig	N/A	N/A
5255211	12/1992	Redmond	N/A	N/A
5446833	12/1994	Miller et al.	N/A	N/A
5625765	12/1996	Ellenby et al.	N/A	N/A
5682332	12/1996	Ellenby et al.	N/A	N/A
5742521	12/1997	Ellenby et al.	N/A	N/A
5748194	12/1997	Chen	N/A	N/A
5751576	12/1997	Monson	N/A	N/A
5759044	12/1997	Redmond	N/A	N/A
5815411	12/1997	Ellenby et al.	N/A	N/A
5848373	12/1997	Delorme et al.	N/A	N/A
5884029	12/1998	Brush, II et al.	N/A	N/A
5991827	12/1998	Ellenby et al.	N/A	N/A
6031545	12/1999	Ellenby et al.	N/A	N/A
6037936	12/1999	Ellenby et al.	N/A	N/A
6052125	12/1999	Gardiner et al.	N/A	N/A
6064398	12/1999	Ellenby et al.	N/A	N/A
6064749	12/1999	Hirota et al.	N/A	N/A
6081278	12/1999	Chen	N/A	N/A
6092107	12/1999	Eleftheriadis et al.	N/A	N/A
6097393	12/1999	Prouty, IV et al.	N/A	N/A
6098118	12/1999	Ellenby et al.	N/A	N/A
6130673	12/1999	Pulli et al.	N/A	N/A
6161126	12/1999	Wies et al.	N/A	N/A
6169545	12/2000	Gallery et al.	N/A	N/A
6173239	12/2000	Ellenby	N/A	N/A
6215498	12/2000	Filo et al.	N/A	N/A
6226669	12/2000	Huang et al.	N/A	N/A
6240360	12/2000	Phelan	N/A	N/A
6242944	12/2000	Benedetti et al.	N/A	N/A
6256043	12/2000	Aho et al.	N/A	N/A
6278461	12/2000	Ellenby et al.	N/A	N/A
6307556	12/2000	Ellenby et al.	N/A	N/A
6308565	12/2000	French et al.	N/A	N/A
6336098	12/2001	Fortenberry et al.	N/A	N/A
6339745	12/2001	Novik et al.	N/A	N/A
6346938	12/2001	Chan et al.	N/A	N/A
6396475	12/2001	Ellenby et al.	N/A	N/A
6414696	12/2001	Ellenby et al.	N/A	N/A
6512844	12/2002	Bouguet et al.	N/A	N/A
6522292	12/2002	Ellenby et al.	N/A	N/A
6529331	12/2002	Massof et al.	N/A	N/A

6535210	12/2002	Ellenby et al.	N/A	N/A
6552729	12/2002	Bernardo et al.	N/A	N/A
6552744	12/2002	Chen	N/A	N/A
6553310	12/2002	Lopke	N/A	N/A
6557041	12/2002	Mallart	N/A	N/A
6559813	12/2002	DeLuca et al.	N/A	N/A
6563489	12/2002	Latypov et al.	N/A	N/A
6563529	12/2002	Jongerius	N/A	N/A
6577714	12/2002	Darcie et al.	N/A	N/A
6631403	12/2002	Deutsch et al.	N/A	N/A
6672961	12/2003	Uzun	N/A	N/A
6690370	12/2003	Ellenby et al.	N/A	N/A
6691032	12/2003	Irish et al.	N/A	N/A
6746332	12/2003	Ing et al.	N/A	N/A
6751655	12/2003	Deutsch et al.	N/A	N/A
6757068	12/2003	Foxlin	N/A	N/A
6767287	12/2003	Mcquaid et al.	N/A	N/A
6768509	12/2003	Bradski et al.	N/A	N/A
6774869	12/2003	Biocca et al.	N/A	N/A
6785667	12/2003	Orbanes et al.	N/A	N/A
6804726	12/2003	Ellenby et al.	N/A	N/A
6822648	12/2003	Furlong et al.	N/A	N/A
6853398	12/2004	Malzbender et al.	N/A	N/A
6854012	12/2004	Taylor	N/A	N/A
6882933	12/2004	Kondou et al.	N/A	N/A
6922155	12/2004	Evans et al.	N/A	N/A
6930715	12/2004	Mower	N/A	N/A
6965371	12/2004	MacLean et al.	N/A	N/A
6968973	12/2004	Uyttendaele et al.	N/A	N/A
7016532	12/2005	Boncyk et al.	N/A	N/A
7031875	12/2005	Ellenby et al.	N/A	N/A
7073129	12/2005	Robarts et al.	N/A	N/A
7076505	12/2005	Campbell	N/A	N/A
7113618	12/2005	Junkins et al.	N/A	N/A
7116326	12/2005	Soulchin et al.	N/A	N/A
7116342	12/2005	Dengler et al.	N/A	N/A
7142209	12/2005	Uyttendaele et al.	N/A	N/A
7143258	12/2005	Bae	N/A	N/A
7168042	12/2006	Braun et al.	N/A	N/A
7174301	12/2006	Florance et al.	N/A	N/A
7206000	12/2006	Zitnick, III et al.	N/A	N/A
7245273	12/2006	Eberl et al.	N/A	N/A
7269425	12/2006	Valkóet al.	N/A	N/A
7271795	12/2006	Bradski	N/A	N/A
7274380	12/2006	Navab et al.	N/A	N/A
7280697	12/2006	Perona et al.	N/A	N/A
7301536	12/2006	Ellenby et al.	N/A	N/A
7353114	12/2007	Rohlf et al.	N/A	N/A
7369668	12/2007	Huopaniemi et al.	N/A	N/A
7395507	12/2007	Robarts et al.	N/A	N/A
7406421	12/2007	Odinak et al.	N/A	N/A
7412427	12/2007	Zitnick et al.	N/A	N/A
7454361	12/2007	Halavais et al.	N/A	N/A
7477780	12/2008	Boncyk et al.	N/A	N/A
7511736	12/2008	Benton	N/A	N/A
7529639	12/2008	Rasanen et al.	N/A	N/A
7532224	12/2008	Bannai	N/A	N/A

7564469	12/2008	Cohen	N/A	N/A
7565008	12/2008	Boncyk et al.	N/A	N/A
7641342	12/2009	Eberl et al.	N/A	N/A
7650616	12/2009	Lee	N/A	N/A
7680324	12/2009	Boncyk et al.	N/A	N/A
7696905	12/2009	Ellenby et al.	N/A	N/A
7710395	12/2009	Rodgers et al.	N/A	N/A
7714895	12/2009	Pretlove et al.	N/A	N/A
7729946	12/2009	Chu	N/A	N/A
7734412	12/2009	Shi et al.	N/A	N/A
7768534	12/2009	Pentenrieder et al.	N/A	N/A
7774180	12/2009	Joussemet et al.	N/A	N/A
7796155	12/2009	Neely, III et al.	N/A	N/A
7817104	12/2009	Ryu et al.	N/A	N/A
7822539	12/2009	Akiyoshi et al.	N/A	N/A
7828655	12/2009	Uhlir et al.	N/A	N/A
7844229	12/2009	Gyorfi et al.	N/A	N/A
7847699	12/2009	Lee et al.	N/A	N/A
7847808	12/2009	Cheng et al.	N/A	N/A
7887421	12/2010	Tabata	N/A	N/A
7889193	12/2010	Platonov et al.	N/A	N/A
7899915	12/2010	Reisman	N/A	N/A
7904577	12/2010	Taylor	N/A	N/A
7907128	12/2010	Bathiche et al.	N/A	N/A
7908462	12/2010	Sung	N/A	N/A
7916138	12/2010	John et al.	N/A	N/A
7962281	12/2010	Rasmussen et al.	N/A	N/A
7978207	12/2010	Herf et al.	N/A	N/A
8046408	12/2010	Torabi	N/A	N/A
8118297	12/2011	Izumichi	N/A	N/A
8130242	12/2011	Cohen	N/A	N/A
8130260	12/2011	Krill et al.	N/A	N/A
8160994	12/2011	Ong et al.	N/A	N/A
8170222	12/2011	Dunko	N/A	N/A
8189959	12/2011	Szeliski et al.	N/A	N/A
8190749	12/2011	Chi et al.	N/A	N/A
8204299	12/2011	Arcas et al.	N/A	N/A
8218873	12/2011	Boncyk et al.	N/A	N/A
8223024	12/2011	Petrou et al.	N/A	N/A
8223088	12/2011	Gomez et al.	N/A	N/A
8224077	12/2011	Boncyk et al.	N/A	N/A
8224078	12/2011	Boncyk et al.	N/A	N/A
8246467	12/2011	Huang et al.	N/A	N/A
8251819	12/2011	Watkins, Jr. et al.	N/A	N/A
8291346	12/2011	Kerr et al.	N/A	N/A
8315432	12/2011	Lefevre et al.	N/A	N/A
8321527	12/2011	Martin et al.	N/A	N/A
8374395	12/2012	Lefevre et al.	N/A	N/A
8417261	12/2012	Huston et al.	N/A	N/A
8427508	12/2012	Mattila et al.	N/A	N/A
8438110	12/2012	Calman et al.	N/A	N/A
8472972	12/2012	Nadler et al.	N/A	N/A
8488011	12/2012	Blanchflower et al.	N/A	N/A
8489993	12/2012	Tamura et al.	N/A	N/A
8498814	12/2012	Irish et al.	N/A	N/A
8502835	12/2012	Meehan	N/A	N/A
8509483	12/2012	Inigo	N/A	N/A

8519844	12/2012	Richey et al.	N/A	N/A
8527340	12/2012	Fisher et al.	N/A	N/A
8531449	12/2012	Lynch et al.	N/A	N/A
8537113	12/2012	Weising et al.	N/A	N/A
8558759	12/2012	Gomez et al.	N/A	N/A
8576276	12/2012	Bar-Zeev et al.	N/A	N/A
8576756	12/2012	Ko et al.	N/A	N/A
8585476	12/2012	Mullen et al.	N/A	N/A
8605141	12/2012	Dialameh et al.	N/A	N/A
8606657	12/2012	Chesnut et al.	N/A	N/A
8633946	12/2013	Cohen	N/A	N/A
8645220	12/2013	Harper et al.	N/A	N/A
8660369	12/2013	Llano et al.	N/A	N/A
8660951	12/2013	Calman et al.	N/A	N/A
8675017	12/2013	Rose et al.	N/A	N/A
8686924	12/2013	Braun et al.	N/A	N/A
8700060	12/2013	Huang	N/A	N/A
8706170	12/2013	Jacobsen et al.	N/A	N/A
8706399	12/2013	Irish et al.	N/A	N/A
8711176	12/2013	Douris et al.	N/A	N/A
8727887	12/2013	Mahajan et al.	N/A	N/A
8730156	12/2013	Weising et al.	N/A	N/A
8743145	12/2013	Price et al.	N/A	N/A
8743244	12/2013	Vartanian et al.	N/A	N/A
8744214	12/2013	Snavely et al.	N/A	N/A
8745494	12/2013	Spivack	N/A	N/A
8751159	12/2013	Hall	N/A	N/A
8754907	12/2013	Tseng	N/A	N/A
8762047	12/2013	Sterkel et al.	N/A	N/A
8764563	12/2013	Toyoda	N/A	N/A
8786675	12/2013	Deering et al.	N/A	N/A
8803917	12/2013	Meehan	N/A	N/A
8810598	12/2013	Soon-Shiong	N/A	N/A
8814691	12/2013	Haddick et al.	N/A	N/A
8855719	12/2013	Jacobsen et al.	N/A	N/A
8872851	12/2013	Choubassi et al.	N/A	N/A
8893164	12/2013	Teller	N/A	N/A
8913085	12/2013	Anderson et al.	N/A	N/A
8933841	12/2014	Valaee et al.	N/A	N/A
8938464	12/2014	Bailly et al.	N/A	N/A
8958979	12/2014	Levine et al.	N/A	N/A
8965741	12/2014	McCulloch et al.	N/A	N/A
8968099	12/2014	Hanke et al.	N/A	N/A
8994645	12/2014	Meehan	N/A	N/A
9001252	12/2014	Hannaford	N/A	N/A
9007364	12/2014	Bailey	N/A	N/A
9024842	12/2014	Gomez et al.	N/A	N/A
9024972	12/2014	Bronder et al.	N/A	N/A
9026940	12/2014	Jung et al.	N/A	N/A
9037468	12/2014	Osman	N/A	N/A
9041739	12/2014	Latta et al.	N/A	N/A
9047609	12/2014	Ellis et al.	N/A	N/A
9071709	12/2014	Wither et al.	N/A	N/A
9098905	12/2014	Rivlin et al.	N/A	N/A
9122053	12/2014	Geisner et al.	N/A	N/A
9122321	12/2014	Perez et al.	N/A	N/A
9122368	12/2014	Szeliski et al.	N/A	N/A

9122707	12/2014	Wither et al.	N/A	N/A
9128520	12/2014	Geisner et al.	N/A	N/A
9129644	12/2014	Gay et al.	N/A	N/A
9131208	12/2014	Jin	N/A	N/A
9143839	12/2014	Reisman et al.	N/A	N/A
9167386	12/2014	Valaee et al.	N/A	N/A
9177381	12/2014	McKinnon	N/A	N/A
9178953	12/2014	Theimer et al.	N/A	N/A
9182815	12/2014	Small et al.	N/A	N/A
9183560	12/2014	Abelow	N/A	N/A
9230367	12/2015	Stroila	N/A	N/A
9240074	12/2015	Berkovich et al.	N/A	N/A
9245387	12/2015	Poulos et al.	N/A	N/A
9262743	12/2015	Heins et al.	N/A	N/A
9264515	12/2015	Ganapathy et al.	N/A	N/A
9280258	12/2015	Bailly et al.	N/A	N/A
9311397	12/2015	Meadow et al.	N/A	N/A
9317133	12/2015	Korah et al.	N/A	N/A
9345957	12/2015	Geisner et al.	N/A	N/A
9377862	12/2015	Parkinson et al.	N/A	N/A
9384737	12/2015	Lamb et al.	N/A	N/A
9389090	12/2015	Levine et al.	N/A	N/A
9396589	12/2015	Soon-Shiong	N/A	N/A
9466144	12/2015	Sharp et al.	N/A	N/A
9480913	12/2015	Briggs	N/A	N/A
9482528	12/2015	Baker et al.	N/A	N/A
9495591	12/2015	Visser et al.	N/A	N/A
9495760	12/2015	Swaminathan et al.	N/A	N/A
9498720	12/2015	Geisner et al.	N/A	N/A
9503310	12/2015	Hawkes et al.	N/A	N/A
9536251	12/2016	Huang et al.	N/A	N/A
9552673	12/2016	Hilliges et al.	N/A	N/A
9558557	12/2016	Jiang et al.	N/A	N/A
9573064	12/2016	Kinnebrew et al.	N/A	N/A
9582516	12/2016	McKinnon et al.	N/A	N/A
9602859	12/2016	Strong	N/A	N/A
9662582	12/2016	Mullen	N/A	N/A
9678654	12/2016	Wong et al.	N/A	N/A
9782668	12/2016	Golden et al.	N/A	N/A
9805385	12/2016	Soon-Shiong	N/A	N/A
9817848	12/2016	McKinnon et al.	N/A	N/A
9824501	12/2016	Soon-Shiong	N/A	N/A
9891435	12/2017	Boger et al.	N/A	N/A
9942420	12/2017	Rao et al.	N/A	N/A
9972208	12/2017	Levine et al.	N/A	N/A
10002337	12/2017	Siddique et al.	N/A	N/A
10007928	12/2017	Graham et al.	N/A	N/A
10062213	12/2017	Mount et al.	N/A	N/A
10068381	12/2017	Blanchflower et al.	N/A	N/A
10115122	12/2017	Soon-Shiong	N/A	N/A
10127733	12/2017	Soon-Shiong	N/A	N/A
10133342	12/2017	Mittal et al.	N/A	N/A
10140317	12/2017	McKinnon et al.	N/A	N/A
10147113	12/2017	Soon-Shiong	N/A	N/A
10217284	12/2018	Das et al.	N/A	N/A
10304073	12/2018	Soon-Shiong	N/A	N/A
10339717	12/2018	Weisman et al.	N/A	N/A

10403051	12/2018	Soon-Shiong	N/A	N/A
10509461	12/2018	Mullen	N/A	N/A
10565828	12/2019	Amaitis et al.	N/A	N/A
10614477	12/2019	Soon-Shiong	N/A	N/A
10664518	12/2019	McKinnon et al.	N/A	N/A
10675543	12/2019	Reiche, III	N/A	N/A
10828559	12/2019	Mullen	N/A	N/A
10838485	12/2019	Mullen	N/A	N/A
11004102	12/2020	Soon-Shiong	N/A	N/A
11107289	12/2020	Soon-Shiong	N/A	N/A
11263822	12/2021	Weisman et al.	N/A	N/A
11270114	12/2021	Park et al.	N/A	N/A
11514652	12/2021	Soon-Shiong	N/A	N/A
11521226	12/2021	Soon-Shiong	N/A	N/A
11645668	12/2022	Soon-Shiong	N/A	N/A
11854036	12/2022	Soon-Shiong	N/A	N/A
11854153	12/2022	Soon-Shiong	N/A	N/A
11869160	12/2023	Soon-Shiong	N/A	N/A
2001/0045978	12/2000	McConnell et al.	N/A	N/A
2002/0044152	12/2001	Abbott, III et al.	N/A	N/A
2002/0077905	12/2001	Arndt et al.	N/A	N/A
2002/0080167	12/2001	Andrews et al.	N/A	N/A
2002/0086669	12/2001	Bos et al.	N/A	N/A
2002/0107634	12/2001	Luciani	N/A	N/A
2002/0133291	12/2001	Hamada et al.	N/A	N/A
2002/0138607	12/2001	Rourke et al.	N/A	N/A
2002/0158873	12/2001	Williamson	N/A	N/A
2002/0163521	12/2001	Ellenby et al.	N/A	N/A
2003/0004802	12/2002	Callegari	N/A	N/A
2003/0008619	12/2002	Werner	N/A	N/A
2003/0027634	12/2002	Matthews, III	N/A	N/A
2003/0060211	12/2002	Chern et al.	N/A	N/A
2003/0069693	12/2002	Snapp et al.	N/A	N/A
2003/0177187	12/2002	Levine et al.	N/A	N/A
2003/0195022	12/2002	Lynch et al.	N/A	N/A
2003/0212996	12/2002	Wolzien	N/A	N/A
2003/0224855	12/2002	Cunningham	N/A	N/A
2003/0234859	12/2002	Malzbender et al.	N/A	N/A
2004/0002843	12/2003	Robarts et al.	N/A	N/A
2004/0058732	12/2003	Piccionelli	N/A	N/A
2004/0104935	12/2003	Williamson et al.	N/A	N/A
2004/0110565	12/2003	Levesque	N/A	N/A
2004/0164897	12/2003	Treadwell et al.	N/A	N/A
2004/0193441	12/2003	Altieri	N/A	N/A
2004/0203380	12/2003	Hamdi et al.	N/A	N/A
2004/0221053	12/2003	Codella et al.	N/A	N/A
2004/0223190	12/2003	Oka	N/A	N/A
2004/0246333	12/2003	Steuart, III	N/A	N/A
2004/0248653	12/2003	Barros et al.	N/A	N/A
2005/0004753	12/2004	Weiland et al.	N/A	N/A
2005/0024501	12/2004	Ellenby et al.	N/A	N/A
2005/0043097	12/2004	March et al.	N/A	N/A
2005/0047647	12/2004	Rutishauser et al.	N/A	N/A
2005/0049022	12/2004	Mullen	N/A	N/A
2005/0060377	12/2004	Lo et al.	N/A	N/A
2005/0143172	12/2004	Kurzweil	N/A	N/A
2005/0192025	12/2004	Kaplan	N/A	N/A

2005/0197767	12/2004	Nortrup	N/A	N/A
2005/0202877	12/2004	Uhlir et al.	N/A	N/A
2005/0208457	12/2004	Fink et al.	N/A	N/A
2005/0223031	12/2004	Zisserman et al.	N/A	N/A
2005/0285878	12/2004	Singh et al.	N/A	N/A
2005/0289590	12/2004	Cheok et al.	N/A	N/A
2006/0010256	12/2005	Heron et al.	N/A	N/A
2006/0025229	12/2005	Mahajan et al.	N/A	N/A
2006/0038833	12/2005	Mallinson et al.	N/A	N/A
2006/0047704	12/2005	Gopalakrishnan	N/A	N/A
2006/0105838	12/2005	Mullen	N/A	N/A
2006/0160619	12/2005	Skoglund	N/A	N/A
2006/0161379	12/2005	Ellenby et al.	N/A	N/A
2006/0166740	12/2005	Sufuentes	N/A	N/A
2006/0190812	12/2005	Ellenby et al.	N/A	N/A
2006/0223635	12/2005	Rosenberg	N/A	N/A
2006/0223637	12/2005	Rosenberg	N/A	N/A
2006/0249572	12/2005	Chen et al.	N/A	N/A
2006/0259361	12/2005	Barhydt et al.	N/A	N/A
2006/0262140	12/2005	Kujawa et al.	N/A	N/A
2007/0035562	12/2006	Azuma et al.	N/A	N/A
2007/0038944	12/2006	Carignano et al.	N/A	N/A
2007/0060408	12/2006	Schultz et al.	N/A	N/A
2007/0066358	12/2006	Silverbrook et al.	N/A	N/A
2007/0070069	12/2006	Samarasekera et al.	N/A	N/A
2007/0087828	12/2006	Robertson et al.	N/A	N/A
2007/0099703	12/2006	Terebilo	N/A	N/A
2007/0109619	12/2006	Eberl et al.	N/A	N/A
2007/0146391	12/2006	Pentenrieder et al.	N/A	N/A
2007/0162341	12/2006	McConnell et al.	N/A	N/A
2007/0167237	12/2006	Wang et al.	N/A	N/A
2007/0173265	12/2006	Gum	N/A	N/A
2007/0182739	12/2006	Platonov et al.	N/A	N/A
2007/0265089	12/2006	Robarts et al.	N/A	N/A
2007/0271301	12/2006	Klive	N/A	N/A
2007/0288332	12/2006	Naito	N/A	N/A
2008/0024594	12/2007	Ritchey	N/A	N/A
2008/0030429	12/2007	Hailpern et al.	N/A	N/A
2008/0071559	12/2007	Arrasvuori	N/A	N/A
2008/0081638	12/2007	Boland et al.	N/A	N/A
2008/0106489	12/2007	Brown et al.	N/A	N/A
2008/0125218	12/2007	Collins et al.	N/A	N/A
2008/0129528	12/2007	Guthrie	N/A	N/A
2008/0132251	12/2007	Altman et al.	N/A	N/A
2008/0147325	12/2007	Maassel et al.	N/A	N/A
2008/0154538	12/2007	Stathis	N/A	N/A
2008/0157946	12/2007	Eberl et al.	N/A	N/A
2008/0198159	12/2007	Liu et al.	N/A	N/A
2008/0198222	12/2007	Gowda	N/A	N/A
2008/0211813	12/2007	Jamwal et al.	N/A	N/A
2008/0261697	12/2007	Chatani et al.	N/A	N/A
2008/0262910	12/2007	Altberg et al.	N/A	N/A
2008/0268876	12/2007	Gelfand et al.	N/A	N/A
2008/0291205	12/2007	Rasmussen et al.	N/A	N/A
2008/0319656	12/2007	Irish	N/A	N/A
2009/0003662	12/2008	Joseph et al.	N/A	N/A
2009/0013052	12/2008	Robarts et al.	N/A	N/A

2009/0037103	12/2008	Herbst et al.	N/A	N/A
2009/0061901	12/2008	ArrasVuori et al.	N/A	N/A
2009/0081959	12/2008	Gyorfi et al.	N/A	N/A
2009/0102859	12/2008	Athsani et al.	N/A	N/A
2009/0144148	12/2008	Jung et al.	N/A	N/A
2009/0149250	12/2008	Middleton	N/A	N/A
2009/0167787	12/2008	Bathiche et al.	N/A	N/A
2009/0167919	12/2008	Anttila et al.	N/A	N/A
2009/0176509	12/2008	Davis et al.	N/A	N/A
2009/0187389	12/2008	Dobbins et al.	N/A	N/A
2009/0193055	12/2008	Kuberka et al.	N/A	N/A
2009/0195650	12/2008	Hanai et al.	N/A	N/A
2009/0209270	12/2008	Gutierrez et al.	N/A	N/A
2009/0210486	12/2008	Lim	N/A	N/A
2009/0213114	12/2008	Dobbins et al.	N/A	N/A
2009/0219224	12/2008	Elg et al.	N/A	N/A
2009/0222742	12/2008	Pelton et al.	N/A	N/A
2009/0237546	12/2008	Bloebaum et al.	N/A	N/A
2009/0248300	12/2008	Dunko et al.	N/A	N/A
2009/0271160	12/2008	Copenhagen et al.	N/A	N/A
2009/0271715	12/2008	Tumuluri	N/A	N/A
2009/0284553	12/2008	Seydoux	N/A	N/A
2009/0285483	12/2008	Guven et al.	N/A	N/A
2009/0287587	12/2008	Bloebaum et al.	N/A	N/A
2009/0289956	12/2008	Douris et al.	N/A	N/A
2009/0293012	12/2008	Alter et al.	N/A	N/A
2009/0319902	12/2008	Kneller et al.	N/A	N/A
2009/0322671	12/2008	Scott et al.	N/A	N/A
2009/0325607	12/2008	Conway et al.	N/A	N/A
2010/0008255	12/2009	Khosravy et al.	N/A	N/A
2010/0023878	12/2009	Douris et al.	N/A	N/A
2010/0045933	12/2009	Eberl et al.	N/A	N/A
2010/0048242	12/2009	Rhoads et al.	N/A	N/A
2010/0087250	12/2009	Chiu	N/A	N/A
2010/0113157	12/2009	Chin et al.	N/A	N/A
2010/0138294	12/2009	Bussmann et al.	N/A	N/A
2010/0162149	12/2009	Sheleheda et al.	N/A	N/A
2010/0185504	12/2009	Rajan et al.	N/A	N/A
2010/0188638	12/2009	Eberl et al.	N/A	N/A
2010/0189309	12/2009	Rouzes et al.	N/A	N/A
2010/0194782	12/2009	Gyorfi et al.	N/A	N/A
2010/0208033	12/2009	Edge et al.	N/A	N/A
2010/0211506	12/2009	Chang et al.	N/A	N/A
2010/0217855	12/2009	Przybysz et al.	N/A	N/A
2010/0241628	12/2009	Levanon	N/A	N/A
2010/0246969	12/2009	Winder et al.	N/A	N/A
2010/0257252	12/2009	Dougherty et al.	N/A	N/A
2010/0287485	12/2009	Bertolami et al.	N/A	N/A
2010/0302143	12/2009	Spivack	N/A	N/A
2010/0306120	12/2009	Ciptawilangga	N/A	N/A
2010/0309097	12/2009	Raviv et al.	N/A	N/A
2010/0315418	12/2009	Woo	N/A	N/A
2010/0321389	12/2009	Gay et al.	N/A	N/A
2010/0321540	12/2009	Woo et al.	N/A	N/A
2010/0325154	12/2009	Schloter et al.	N/A	N/A
2011/0018903	12/2010	Lapstun et al.	N/A	N/A
2011/0028220	12/2010	Reiche, III	N/A	N/A

2011/0034176	12/2010	Lord et al.	N/A	N/A
2011/0038634	12/2010	DeCusatis et al.	N/A	N/A
2011/0039622	12/2010	Levenson et al.	N/A	N/A
2011/0055049	12/2010	Harper et al.	N/A	N/A
2011/0093326	12/2010	Bous et al.	N/A	N/A
2011/0134108	12/2010	Hertenstein	N/A	N/A
2011/0142016	12/2010	Chatterjee	N/A	N/A
2011/0145051	12/2010	Paradise et al.	N/A	N/A
2011/0148922	12/2010	Son et al.	N/A	N/A
2011/0151955	12/2010	Nave	N/A	N/A
2011/0153186	12/2010	Jakobson	N/A	N/A
2011/0183754	12/2010	Alghamdi	N/A	N/A
2011/0202460	12/2010	Buer et al.	N/A	N/A
2011/0205242	12/2010	Friesen	N/A	N/A
2011/0212762	12/2010	Ocko et al.	N/A	N/A
2011/0216060	12/2010	Weising et al.	N/A	N/A
2011/0221771	12/2010	Cramer et al.	N/A	N/A
2011/0225069	12/2010	Cramer et al.	N/A	N/A
2011/0234631	12/2010	Kim et al.	N/A	N/A
2011/0238751	12/2010	Belimpasakis et al.	N/A	N/A
2011/0241976	12/2010	Boger et al.	N/A	N/A
2011/0246064	12/2010	Nicholson	N/A	N/A
2011/0246276	12/2010	Peters et al.	N/A	N/A
2011/0249122	12/2010	Tricoukes et al.	N/A	N/A
2011/0279445	12/2010	Murphy et al.	N/A	N/A
2011/0282747	12/2010	Lavrov et al.	N/A	N/A
2011/0316880	12/2010	Ojala et al.	N/A	N/A
2011/0319148	12/2010	Kinnebrew et al.	N/A	N/A
2012/0019557	12/2011	Aronsson et al.	N/A	N/A
2012/0050144	12/2011	Morlock	N/A	N/A
2012/0050503	12/2011	Kraft	N/A	N/A
2012/0075342	12/2011	Choubassi	345/633	G06T 17/00
2012/0092328	12/2011	Flaks et al.	N/A	N/A
2012/0098859	12/2011	Lee et al.	N/A	N/A
2012/0105473	12/2011	Bar-Zeev et al.	N/A	N/A
2012/0105474	12/2011	Cudalbu et al.	N/A	N/A
2012/0105475	12/2011	Tseng et al.	N/A	N/A
2012/0109773	12/2011	Sipper et al.	N/A	N/A
2012/0110477	12/2011	Gaume	N/A	N/A
2012/0113141	12/2011	Zimmerman et al.	N/A	N/A
2012/0116920	12/2011	Adhikari et al.	N/A	N/A
2012/0122570	12/2011	Baronoff	N/A	N/A
2012/0127062	12/2011	Bar-Zeev et al.	N/A	N/A
2012/0127201	12/2011	Kim et al.	N/A	N/A
2012/0127284	12/2011	Bar-Zeev et al.	N/A	N/A
2012/0139817	12/2011	Freeman	N/A	N/A
2012/0150746	12/2011	Graham	N/A	N/A
2012/0157210	12/2011	Hall	N/A	N/A
2012/0162255	12/2011	Ganapathy et al.	N/A	N/A
2012/0194547	12/2011	Johnson et al.	N/A	N/A
2012/0206452	12/2011	Geisner et al.	N/A	N/A
2012/0219181	12/2011	Tseng et al.	N/A	N/A
2012/0226437	12/2011	Li et al.	N/A	N/A
2012/0229625	12/2011	Calman et al.	N/A	N/A
2012/0231424	12/2011	Calman et al.	N/A	N/A
2012/0231891	12/2011	Watkins, Jr. et al.	N/A	N/A
2012/0232968	12/2011	Calman et al.	N/A	N/A

2012/0232976	12/2011	Calman et al.	N/A	N/A
2012/0236025	12/2011	Jacobsen et al.	N/A	N/A
2012/0244950	12/2011	Braun	N/A	N/A
2012/0252359	12/2011	Adams et al.	N/A	N/A
2012/0256917	12/2011	Lieberman et al.	N/A	N/A
2012/0260538	12/2011	Schob et al.	N/A	N/A
2012/0276997	12/2011	Chowdhary et al.	N/A	N/A
2012/0287284	12/2011	Jacobsen et al.	N/A	N/A
2012/0293506	12/2011	Vertucci et al.	N/A	N/A
2012/0302129	12/2011	Persaud et al.	N/A	N/A
2013/0021373	12/2012	Vaught et al.	N/A	N/A
2013/0044042	12/2012	Olsson et al.	N/A	N/A
2013/0044128	12/2012	Liu et al.	N/A	N/A
2013/0050258	12/2012	Liu et al.	N/A	N/A
2013/0050496	12/2012	Jeong	N/A	N/A
2013/0064426	12/2012	Watkins, Jr. et al.	N/A	N/A
2013/0073988	12/2012	Groten et al.	N/A	N/A
2013/0076788	12/2012	Zvi	N/A	N/A
2013/0124563	12/2012	CaveLie et al.	N/A	N/A
2013/0128060	12/2012	Rhoads et al.	N/A	N/A
2013/0141419	12/2012	Mount et al.	N/A	N/A
2013/0147836	12/2012	Small et al.	N/A	N/A
2013/0159096	12/2012	Santhanagopal et al.	N/A	N/A
2013/0176202	12/2012	Gervautz	N/A	N/A
2013/0178257	12/2012	Langseth	N/A	N/A
2013/0236040	12/2012	Crawford et al.	N/A	N/A
2013/0326364	12/2012	Latta et al.	N/A	N/A
2013/0335405	12/2012	Scavezze et al.	N/A	N/A
2013/0342572	12/2012	Poulos et al.	N/A	N/A
2014/0002492	12/2013	Lamb et al.	N/A	N/A
2014/0101608	12/2013	Ryskamp et al.	N/A	N/A
2014/0161323	12/2013	Livyatan et al.	N/A	N/A
2014/0168261	12/2013	Margolis et al.	N/A	N/A
2014/0184749	12/2013	Hilliges et al.	N/A	N/A
2014/0267234	12/2013	Hook	345/419	G06T 19/006
2014/0306866	12/2013	Miller et al.	N/A	N/A
2015/0091941	12/2014	Das et al.	N/A	N/A
2015/0172626	12/2014	Martini	N/A	N/A
2015/0206349	12/2014	Rosenthal et al.	N/A	N/A
2015/0288944	12/2014	Nistico et al.	N/A	N/A
2016/0269712	12/2015	Ostrover et al.	N/A	N/A
2016/0292924	12/2015	Balachandreswaran et al.	N/A	N/A
2017/0045941	12/2016	Tokubo et al.	N/A	N/A
2017/0087465	12/2016	Lyons et al.	N/A	N/A
2017/0216099	12/2016	Saladino	N/A	N/A
2018/0300822	12/2017	Papakipos et al.	N/A	N/A
2020/0005547	12/2019	Soon-Shiong	N/A	N/A
2020/0257721	12/2019	McKinnon et al.	N/A	N/A
2021/0358223	12/2020	Soon-Shiong	N/A	N/A
2022/0156314	12/2021	McKinnon et al.	N/A	N/A
2023/0051746	12/2022	Soon-Shiong	N/A	N/A
2024/0062236	12/2023	Soon-Shiong	N/A	N/A
2024/0062486	12/2023	Soon-Shiong	N/A	N/A
2024/0212297	12/2023	Soon-Shiong	N/A	N/A
2025/0095303	12/2024	Soon-Shiong	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
------------	------------------	---------	-----

2 311 319	12/1998	CA	N/A
2235030	12/1998	CA	N/A
2233047	12/1999	CA	N/A
102436461	12/2011	CN	N/A
102484730	12/2011	CN	N/A
102509342	12/2011	CN	N/A
102509348	12/2011	CN	N/A
1 012 725	12/1999	EP	N/A
246 080	12/2001	EP	N/A
1 354 260	12/2002	EP	N/A
119 798	12/2004	EP	N/A
1 965 344	12/2007	EP	N/A
2 207 113	12/2009	EP	N/A
1 588 537	12/2009	EP	N/A
2484384	12/2011	GB	N/A
2001-286674	12/2000	JP	N/A
2002-056163	12/2001	JP	N/A
2002-282553	12/2001	JP	N/A
2002-346226	12/2001	JP	N/A
2003-305276	12/2002	JP	N/A
2003-337903	12/2002	JP	N/A
2004-64398	12/2003	JP	N/A
2004-078385	12/2003	JP	N/A
2005-196494	12/2004	JP	N/A
2005-215922	12/2004	JP	N/A
2005-316977	12/2004	JP	N/A
2006-085518	12/2005	JP	N/A
2006-190099	12/2005	JP	N/A
2006-280480	12/2005	JP	N/A
2007-222640	12/2006	JP	N/A
2010-102588	12/2009	JP	N/A
2010-118019	12/2009	JP	N/A
2010-224884	12/2009	JP	N/A
2011-60254	12/2010	JP	N/A
2011-153324	12/2010	JP	N/A
2011-253324	12/2010	JP	N/A
2012-014220	12/2011	JP	N/A
2010-0124947	12/2009	KR	N/A
20120082672	12/2011	KR	N/A
10-1171264	12/2011	KR	N/A
95/09411	12/1994	WO	N/A
97/44737	12/1996	WO	N/A
98/50884	12/1997	WO	N/A
99/42946	12/1998	WO	N/A
99/42947	12/1998	WO	N/A
00/20929	12/1999	WO	N/A
01/63487	12/2000	WO	N/A
01/71282	12/2000	WO	N/A
01/88679	12/2000	WO	N/A
02/03091	12/2001	WO	N/A
02/42921	12/2001	WO	N/A
02/059716	12/2001	WO	N/A
02/073818	12/2001	WO	N/A
2007/140155	12/2006	WO	N/A
2010/079876	12/2009	WO	N/A
2010/138344	12/2009	WO	N/A
2011/028720	12/2010	WO	N/A

2011/084720	12/2010	WO	N/A
2011/163063	12/2010	WO	N/A
2012/082807	12/2011	WO	N/A
2012164155	12/2011	WO	N/A
2013/023705	12/2012	WO	N/A
2013095383	12/2012	WO	N/A
2014/108799	12/2013	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion issued in International Application No. PCT/US2012/032204 dated Oct. 29, 2012. cited by applicant

Wauters, "Stanford Graduates Release Pulse, A Must-Have News App for the iPad," Techcrunch.com, techcrunch.com/2010/05/31/pulse-ipad/, 2010. cited by applicant

Hickins, "A License to Pry," The Wall Street Journal, <http://blogs.wsj.com/digits/2011/03/10/a-license-to-pry/tab/print/>, 2011. cited by applicant

Notice of Reasons for Rejection issued in Japanese Patent Application No. 2014-503962 dated Sep. 22, 2014. cited by applicant

Notice of Reasons for Rejection issued in Japanese Patent Application No. 2014-503962 dated Jun. 30, 2015. cited by applicant

European Search Report issued in European Patent Application No. 12767566.8 dated Mar. 20, 2015. cited by applicant
 "3D Laser Mapping Launches Mobile Indoor Mapping System," 3D Laser Mapping, Dec. 3, 2012, 1 page. cited by applicant

Banwell et al., "Combining Absolute Positioning and Vision for Wide Area Augmented Reality," Proceedings of the International Conference on Computer Graphics Theory and Applications, 2010, 4 pages. cited by applicant

Li et al., "3-D Motion Estimation and Online Temporal Calibration for Camera-IMU Systems," Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2013, 8 pages. cited by applicant

Li et al., "High-fidelity Sensor Modeling and Self-Calibration in Vision-aided Inertial Navigation," Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2014, 8 pages. cited by applicant

Li et al., "Online Temporal Calibration for Camera-IMU Systems: Theory and Algorithms," International Journal of Robotics Research, vol. 33, Issue 7, 2014, 16 pages. cited by applicant

Li et al., "Real-time Motion Tracking on a Cellphone using Inertial Sensing and a Rolling-Shutter Camera," Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2013, 8 pages. cited by applicant

Mourikis, "Method for Processing Feature Measurements in Vision-Aided Inertial Navigation," 3 pages, 2013. cited by applicant

Mourikis et al., "Methods for Motion Estimation With a Rolling-Shutter Camera," Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), Karlsruhe, Germany May 6-10, 2013, 9 pages. cited by applicant

Panzarino, "What Exactly Wi-FiSlam Is, and Why Apple Acquired It," <http://thenextweb.com/apple/2013/03/26/what-exactly-wifislam-is-and-why-apple-acquired-it>, Mar. 26, 2013, 10 pages. cited by applicant

Vondrick et al., "HOGgles: Visualizing Object Detection Features," IEEE International Conference on Computer Vision (ICCV), 2013, 9 pages. cited by applicant

Vu et al., "High Accuracy and Visibility-Consistent Dense Multiview Stereo," IEEE Transactions on Pattern Analysis and Machine Intelligence, 2012, vol. 34, No. 5, 13 pages. cited by applicant

International Search Report and Written Opinion issued in International Application No. PCT/US2014/061283 dated Aug. 5, 2015, 11 pages. cited by applicant

Pang et al., "Development of a Process-Based Model for Dynamic Interaction in Spatio-Temporal GIS", GeoInformatica 2002, vol. 6, No. 4, pp. 323-344. cited by applicant

Zhu et al., "The Geometrical Properties of Irregular 2D Voronoi Tessellations," Philosophical Magazine A, 2001, vol. 81, No. 12, pp. 2765-2783. cited by applicant

"S2 Cells," S2Geometry, https://s2geometry.io/devguide/s2cell_hierarchy, 27 pages, Oct. 10, 2019. cited by applicant

Bimber et al., "A Brief Introduction to Augmented Reality, in Spatial Augmented Reality," 2005, CRC Press, 23 pages. cited by applicant

Milgram et al., "A Taxonomy of Mixed Reality Visual Displays," IEEE Transactions on Information and Systems, 1994, vol. 77, No. 12, pp. 1321-1329. cited by applicant

Normand et al., "A new typology of augmented reality applications," Proceedings of the 3rd augmented human international conference, 2012, 9 pages. cited by applicant

Sutherland, "A head-mounted three dimensional display," Proceedings of the Dec. 9-11, 1968, Fall Joint Computer Conference, part I, 1968, pp. 757-764. cited by applicant

Maubon, "A little bit of history from 2006: Nokia's MARA project," <https://www.augmented-reality.fr/2009/03/un-petit-peu-dhistoire-de-2006-projet-mara-de-nokia/>, 7 pages, 2009. cited by applicant

Madden, "Professional Augmented Reality Browsers for Smartphones," 2011, John Wiley & Sons, 44 pages. cited by applicant

Raper et al., "Applications of location-based services: a selected review," *Journal of Location Based Services*, 2007, vol. 1, No. 2, pp. 89-111. cited by applicant

Savage, "Blazing gyros: The evolution of strapdown inertial navigation technology for aircraft," *Journal of Guidance, Control, and Dynamics*, 2013, vol. 36, No. 3, pp. 637-655. cited by applicant

Kim et al., "A Step, Stride and Heading Determination for the Pedestrian Navigation System," *Journal of Global Positioning Systems*, 2004, vol. 3, No. 1-2, pp. 273-279. cited by applicant

"Apple Reinvents the Phone with iPhone," Apple, dated Jan. 9, 2007, <https://www.apple.com/newsroom/2007/01/09Apple-Reinvents-the-Phone-with-iPhone/>, 5 pages. cited by applicant

Macedonia et al., "Exploiting reality with multicast groups: a network architecture for large-scale virtual environments," *Proceedings Virtual Reality Annual International Symposium'95*, 1995, pp. 2-10. cited by applicant

Magerkurth et al., "Pervasive Games: Bringing Computer Entertainment Back to the Real World," *Computers in Entertainment (CIE)*, 2005, vol. 3, No. 3, 19 pages. cited by applicant

Thomas et al., "ARQuake: An Outdoor/Indoor Augmented Reality First Person Application," *Digest of Papers. Fourth International Symposium on Wearable Computers*, 2000, pp. 139-146. cited by applicant

Thomas et al., "First Person Indoor/Outdoor Augmented Reality Application: ARQuake," *Personal and Ubiquitous Computing*, 2002, vol. 6, No. 1, pp. 75-86. cited by applicant

Zyda, "From Visual Simulation to Virtual Reality to Games," *IEEE Computer Society*, 2005, vol. 38, No. 9, pp. 25-32. cited by applicant

Zyda, "Creating a Science of Games," *Communications-ACM*, 2007, vol. 50, No. 7, pp. 26-29. cited by applicant

"Microsoft Computer Dictionary," Microsoft, 2002, 10 pages. cited by applicant

"San Francisco street map," David Rumsey Historical Map Collection, 1953, <https://www.davidrumsey.com/luna/servlet/s/or3ezx>, 2 pages, last accessed 2021. cited by applicant

"Official Transportation Map (2010)," Florida Department of Transportation, https://www.fdot.gov/docs/default-source/geospatial/past_statemap/maps/FLStatemap2010.pdf, 2010, 2 pages. cited by applicant

Krogh, "GPS," *American Society of Media Photographers*, dated Mar. 22, 2010, 10 pages. cited by applicant

"Tigo: Smartphone, 2," *Ads of the World*, https://www.adsoftheworld.com/media/print/tigo_smartphone_2, 6 pages, dated Jan. 2012. cited by applicant

Ta et al., "SURFTrac: Efficient Tracking and Continuous Object Recognition using Local Feature Descriptors," 2009 *IEEE Conference on Computer Vision and Pattern Recognition*, 2009, pp. 2937-2944. cited by applicant

Office Action issued in Chinese Application No. 201710063195.8 dated Mar. 24, 2021, 9 pages. cited by applicant

U.S. Appl. No. 10/438,172 filed May 13, 2003. cited by applicant

U.S. Appl. No. 60/496,752, filed Aug. 21, 2003. cited by applicant

U.S. Appl. No. 60/499,810, filed Sep. 2, 2003. cited by applicant

U.S. Appl. No. 60/502,939, filed Sep. 16, 2003. cited by applicant

U.S. Appl. No. 60/628,475, filed Nov. 16, 2004. cited by applicant

U.S. Appl. No. 61/411,591, filed Nov. 9, 2010. cited by applicant

Vallino, "Interactive Augmented Reality," 1998, University of Rochester, 109 pages. cited by applicant

Jay et al., "Amplifying Head Movements with Head-Mounted Displays," 2003, *Presence*, Massachusetts Institute of Technology, 10 pages. cited by applicant

Julier et al., "Information Filtering for Mobile Augmented Reality," 2000, *IEEE and ACM International Symposium on Augmented Reality*, 10 pages. cited by applicant

Julier et al., "Information Filtering for Mobile Augmented Reality," Jul. 2, 2002, *IEEE*, 6 pages. cited by applicant

Kalwasky, "The Science of Virtual Reality and Virtual Environments," 1993, Addison-Wesley, 215 pages. cited by applicant

Kerr et al., "Wearable Mobile Augmented Reality: Evaluating Outdoor User Experience," 2011, *ACM*, 8 pages. cited by applicant

Kopper et al., "Towards an Understanding of the Effects of Amplified Head Rotations," 2011, *IEEE*, 6 pages. cited by applicant

MacIntyre et al., "Estimating and Adapting to Registration Errors in Augmented Reality Systems," 2002, *Proceedings IEEE Virtual Reality 2002*, 9 pages. cited by applicant

Feißt, "3D Virtual Reality on mobile devices," 2009, VDM Verlag Dr. Muller Aktiengesellschaft & Co. KG, 53 pages. cited by applicant

Chua et al., "MasterMotion: Full Body Wireless Virtual Reality for Tai Chi," Jul. 2002, *ACM SIGGRAPH 2002*

conference abstracts and applications, 1 page. cited by applicant

Melzer et al., “Head-Mounted Displays: Designing for the User,” 2011, 85 pages. cited by applicant

Vorländer, “Auralization—Fundamentals of Acoustics, Modelling, Simulation, Algorithms and Acoustic Virtual Reality,” 2008, Springer, 34 pages. cited by applicant

Miller et al., “The Virtual Museum: Interactive 3D Navigation of a Multimedia Database,” Jul./Sep. 1992, John Wiley & Sons, Ltd., 3 pages. cited by applicant

Koenen, “MPEG-4 Multimedia for our time,” 1999, IEEE Spectrum, 8 pages. cited by applicant

Navab et al., “Laparoscopic Virtual Mirror,” 2007, IEEE Virtual Reality Conference, 8 pages. cited by applicant

Ochi et al., “HMD Viewing Spherical Video Streaming System,” 2014, ACM, 2 pages. cited by applicant

Olson et al., “A Design for a Smartphone-Based Head Mounted Display,” 2011, 2 pages. cited by applicant

Pagarkar et al., “MPEG-4 Tech Paper,” 24 pages, 2002. cited by applicant

Pausch, “Virtual Reality on Five Dollars a Day,” 1991, ACM, 6 pages. cited by applicant

Peternier et al., “Wearable Mixed Reality System in Less Than 1 Pound,” 2006, The Eurographics Assoc. , 10 pages. cited by applicant

Piekarski et al., “ARQuake—Modifications and Hardware for Outdoor Augmented Reality Gaming,” 9 pages, 2003. cited by applicant

Piekarski, “Interactive 3d modelling in outdoor augmented reality worlds,” 2004, The University of South Australia, 264 pages. cited by applicant

Piekarski et al., “Tinmith-Metro: New Outdoor Techniques for Creating City Models with an Augmented Reality Wearable Computer,” 2001, IEEE, 8 pages. cited by applicant

Piekarski et al., “The Tinmith System—Demonstrating New Techniques for Mobile Augmented Reality Modelling,” 2002, 10 pages. cited by applicant

Piekarski et al., “ARQuake: The Outdoor Augmented Reality Gaming System,” 2002, Communications of the ACM, 3 pages. cited by applicant

Piekarski et al., “Integrating Virtual and Augmented Realities in an Outdoor Application,” 1999, 10 pages. cited by applicant

Pimentel et al., “Virtual Reality—Through the new looking glass,” 1993, Windcrest McGraw-Hill , 45 pages. cited by applicant

Basu et al., “Poster: Evolution and Usability of Ubiquitous Immersive 3D Interfaces,” 2013, IEEE, 2 pages. cited by applicant

Pouwelse et al., “A Feasible Low-Power Augmented—Reality Terminal,” 1999, 10 pages. cited by applicant

Madden, “Professional Augmented Reality Browsers for Smartphones,” 2011, John Wiley & Sons, 345 pages. cited by applicant

“Protecting Mobile Privacy: Your Smartphones, Tablets, Cell Phones and Your Privacy—Hearing,” May 10, 2011, U.S. Government Printing Office, 508 pages. cited by applicant

Rashid et al., “Extending Cyberspace: Location Based Games Using Cellular Phones,” 2006, ACM, 18 pages. cited by applicant

Reid et al., “Design for coincidence: Incorporating real world artifacts in location based games,” 2008, ACM, 8 pages. cited by applicant

Rockwell et al., “Campus Mysteries: Serious Walking Around,” 2013, Journal of the Canadian Studies Association, vol. 7(12): 1-18, 18 pages. cited by applicant

Shapiro, “Comparing User Experience in a Panoramic HMD vs. Projection Wall Virtual Reality System,” 2006, Sensics, 12 pages. cited by applicant

Sestito et al., “Intelligent Filtering for Augmented Reality,” 2000, 8 pages. cited by applicant

Sherman et al., “Understanding Virtual Reality,” 2003, Elsevier, 89 pages. cited by applicant

Simcock et al., “Developing a Location Based Tourist Guide Application,” 2003, Australian Computer Society, Inc., 7 pages. cited by applicant

“Sony—Head Mounted Display Reference Guide,” 2011, Sony Corporation, 32 pages. cited by applicant

Hollister, “Sony HMZ-T1 Personal 3D Viewer Review—The Verge, ”Nov. 10, 2011, The Verge, 25 pages. cited by applicant

Gutiérrez et al., “Stepping into Virtual Reality,” 2008, Springer-Verlag, 33 pages. cited by applicant

“Summary of QuickTime for Java,” 90 pages, 1999. cited by applicant

Sutherland, “A head-mounted three dimensional display,” 1968, Fall Joint Computer Conference, 8 pages. cited by applicant

Sutherland, “The Ultimate Display,” 1965, Proceedings of IFIP Congress, 2 pages. cited by applicant

Thomas et al., “ARQuake: An Outdoor/Indoor Augmented Reality First Person Application,” 2000, IEEE, 8 pages. cited by applicant

Thomas et al., "First Person Indoor/Outdoor Augmented Reality Application: ARQuake," 2002, Springer-Verlag, 12 pages. cited by applicant

Wagner et al., "Towards Massively Multi-user Augmented Reality on Handheld Devices," May 2005, Lecture Notes in Computer Science, 13 pages. cited by applicant

Shin et al., "Unified Context-aware Augmented Reality Application Framework for User-Driven Tour Guides," 2010, IEEE, 5 pages. cited by applicant

Julier et al., "Chapter 6—Urban Terrain Modeling for Augmented Reality Applications," 2001, Springer, 20 pages. cited by applicant

"Adobe Flash Video File Format Specification Version 10.1," 2010, Adobe Systems Inc., 89 pages. cited by applicant

Arghire, "Glu Mobile's 1000: Find 'Em All! Game Available in the App Store," <https://news.softpedia.com/news/Glu-Mobile-s-1000-Find-Em-All-Game-Available-in-the-App-Store-134126.shtml>, 2 pages, 2010. cited by applicant

Buchanan, "1,000: Find 'Em All Preview," <https://www.ign.com/articles/2009/10/16/1000-find-em-all-preview>, 8 pages, 2009. cited by applicant

Hirst, "Glu Mobile Announces '1000 Find Em All'. Real World GPS-Based Adventure for iPhone," <https://www.148apps.com/news/glu-mobile-announces-1000-find-em-all-real-world-gpsbased-adventure-iphone/>, 8 pages, 2010. cited by applicant

Tschida, "You Can Now Find 1000: Find 'Em All! In The App Store, <https://appadvice.com/appnn/2010/02/you-can-now-find-1000-find-em-all-in-the-app-store>, 3 pages, 2010. cited by applicant

Piekarski et al., "ARQuake: the outdoor augmented reality gaming system," Communications of the ACM, 2002, vol. 45 No., 1, pp. 36-38. cited by applicant

Piekarski et al., "ARQuake—Modifications and Hardware for Outdoor Augmented Reality Gaming," Linux Australia, 2003, 9 pages. cited by applicant

Randell, "Wearable Computing: A Review," 16 pages, 2005. cited by applicant

Thomas et al., "ARQuake: An Outdoor/Indoor Augmented Reality First Person Application," University of South Australia, 2000, 8 pages. cited by applicant

Thomas et al., "First Person Indoor/Outdoor Augmented Reality Application: ARQuake," Personal and Ubiquitous Computing, 2002, vol. 6, pp. 75-86. cited by applicant

Thomas et al., "Usability and Playability Issues for ARQuake," 2003, 8 pages. cited by applicant

Livingston et al., "An augmented reality system for military operations in urban terrain," Interservice/Industry Training, Simulation, and Education Conference, 2002, vol. 89, 9 pages. cited by applicant

Livingston et al., "Mobile Augmented Reality: Applications and Human Factors Evaluations," 2006, 16 pages. cited by applicant

Cutler, "Dekko Debuts an Augmented Reality Racing Game Playable From the iPad," Techcrunch, <https://techcrunch.com/2013/06/09/dekko-2/>, 8 pages, 2013. cited by applicant

"Dekko's TableTop Speed AR Proof of Concept," www.gametrender.net/2013/06/dekkos-tabletop-speed-ar-proof-of.html, 2 pages, 2013. cited by applicant

"Racing AR Together," <https://augmented.org/2013/06/racing-ar-together/>, 3 pages, 2013. cited by applicant

"Delorme PN-40," www.gpsreview.net/delorme-pn-40/, downloaded on Mar. 8, 2021, 37 pages. cited by applicant

Owings, "DeLorme Earthmate PN-40 review," <https://gpstracklog.com/2009/02/delorme-earthmate-pn-40-review.html>, 17 pages, 2009. cited by applicant

"Astro owner's manual," https://static.garmin.com/pumac//Astro_OwnersManual.pdf, 76 pages, 2009. cited by applicant

"Geko 201 Personal Navigator," https://static.garmin.com/pumac/Geko201_OwnersManual.pdf, 52 pages, 2003. cited by applicant

"GPS 60," https://static.garmincdn.com/pumac/GPS60_OwnersManual.pdf, 90 pages, 2006. cited by applicant

Butler, "How does Google Earth work?," <https://www.nature.com/news/2006/060213/full/060213-7.html>, 2 pages, 2006. cited by applicant

Castello, "How's the weather?," <https://maps.googleblog.com/2007/11/how-s-weather.html>, 3 pages, 2007. cited by applicant

Friedman, "Google Earth for iPhone and iPad," https://www.macworld.com/article/1137794/googleearth_iphone.html, downloaded on Sep. 7, 2010, 3 pages. cited by applicant

"Google Earth," <http://web.archive.org/web/20091213164811/http://earth.google.com/>, 1 page, 2009. cited by applicant

Mellen, "Google Earth 2.0 for iPhone released," https://www.gearthblog.com/blog/archives/2009/11/google_earth_20_for_iphone_released.html, downloaded on Mar. 5, 2021, 5 pages. cited by applicant

"Google Earth iPhone," <http://web.archive.org/web/20091025070614/http://www.google.com/mobile/products/earth.html>, 1 page, 2009. cited by applicant

Shwartz, "Send in the Clouds: Google Earth Adds Weather Layer," <https://searchengineland.com/send-in-the-clouds-google-earth-adds-weather-layer-12651>, 5 pages, 2007. cited by applicant

Senoner, "Google Earth and Microsoft Virtual Earth two Geographic Information Systems," 2007, 44 pages. cited by applicant

Barth, "Official Google Blog: The bright side of sitting in traffic: Crowdsourcing road congestion data," <https://googleblog.blogspot.com/2009/08/bright-side-of-sitting-in-traffic.html>, 4 pages, 2009. cited by applicant

Soni, "Introducing Google Buzz for mobile: See buzz around you and tag posts with your location.," googlemobile.blogspot.com/2010/02/introducing-google-buzz-for-mobile-see.html, 15 pages, 2010. cited by applicant

Chu, "New magical blue circle on your map," <https://googlemobile.blogspot.com/2007/11/new-magical-blue-circle-on-your-map.html>, 20 pages, 2007. cited by applicant

"Google Maps for your phone," <https://web.archive.org/web/20090315195718/http://www.google.com/mobile/default/maps.html>, 2 pages, 2009. cited by applicant

"Get Google on your phone," <http://web.archive.org/web/20091109190817/http://google.com/mobile/#p=default>, 1 page, 2009. cited by applicant

Gundotra, "To 100 million and beyond with Google Maps for mobile," <https://maps.googleblog.com/2010/08/to-100-million-and-beyond-with-google.html>, 6 pages, 2010. cited by applicant

"Introducing Google Buzz for mobile: See buzz around you and tag posts with your location," <https://maps.googleblog.com/2010/02/introducing-google-buzz-for-mobile-see.html>, 16 pages, 2010. cited by applicant

"Google Maps Navigation (Beta)," <http://web.archive.org/web/20091101030954/http://www.google.com:80/mobile/navigation/index.html#p=default>, 3 pages, 2009. cited by applicant

Miller, "Googlepedia: the ultimate Google resource," 2008, Third Edition, 120 pages. cited by applicant

"Upgrade your phone with free Google products," <http://web.archive.org/web/20090315205659/http://www.google.com/mobile/>, 1 page, 2009. cited by applicant

"Google blogging in 2010," <https://googleblog.blogspot.com/2010/>, 50 pages, 2010. cited by applicant

Cheok et al., "Human Pacman: A Mobile Entertainment System with Ubiquitous Computing and Tangible Interaction over a Wide Outdoor Area," 2003, Human-Computer Interaction with Mobile Devices and Services: 5th International Symposium, Mobile HCI 2003, Udine, Italy, Sep. 2003. Proceedings 5, 17 pages. cited by applicant

Knight, "Human PacMan hits real city streets," <https://www.newscientist.com/article/dn6689-human-pacman-hits-real-city-streets/>, 5 pages, 2004. cited by applicant

Sandhana, "Pacman comes to life virtually," <http://news.bbc.co.uk/2/hi/technology/4607449.stm>, 3 pages, 2005. cited by applicant

Wolke, "Digital Wayfinding Apps," <https://web.archive.org/web/20200927073039/https://segd.org/digital-wayfinding-apps>, 5 pages, 2010. cited by applicant

"Nike+ GPS: There's an App for That," <https://www.runnersworld.com/races-places/a20818818/nike-gps-theres-an-app-for-that/>, 3 pages, 2010. cited by applicant

Biggs, "Going the Distance: Nike+ GPS Vs. RunKeeper," <https://techcrunch.com/2010/10/09/going-the-distance-nike-gps-vs-runkeeper/>, 4 pages, 2010. cited by applicant

Rainmaker, "Nike+ Sportwatch GPS in Depth Review," <https://www.dcrainmaker.com/2011/04/nike-sportwatch-gps-in-depth-review.html>, 118 pages, 2011. cited by applicant

Vince, "Introduction to Virtual Reality," 2004, Springer-Verlag, 97 pages. cited by applicant

Fuchs et al., "Virtual Reality: Concepts and Technologies," 2006, CRC Press, 56 pages. cited by applicant

Arieda, "A Virtual / Augmented Reality System with Kinaesthetic Feedback—Virtual Environment with Force Feedback System," 2012, LAP Lambert Academic Publishing, 31 pages. cited by applicant

Sperber et al., "Web-based mobile Augmented Reality: Developing with Layar (3D)," 2010, 7 pages. cited by applicant

"WXHMD—A Wireless Head-Mounted Display with embedded Linux," 2009, Pabr.org, 8 pages. cited by applicant

"XMP Adding Intelligence to Media—XMP Specification Part 3—Storage in Files," 2014, Adobe Systems Inc., 78 pages. cited by applicant

Zhao, "A survey on virtual reality," 2009, Springer, 54 pages. cited by applicant

Zipf et al., "Using Focus Maps to Ease Map Reading—Developing Smart Applications for Mobile Devices," 3 pages, 2002. cited by applicant

Gammeter et al., "Server-side object recognition and client-side object tracking for mobile augmented reality," 2010, IEEE, 8 pages. cited by applicant

Martedi et al., "Foldable Augmented Maps," 2012, IEEE, 11 pages. cited by applicant

Martedi et al., "Foldable Augmented Maps," 2010, IEEE, 8 pages. cited by applicant

Morrison et al., "Like Bees Around the Hive: A Comparative Study of a Mobile Augmented Reality Map," 2009, 10

pages. cited by applicant

Takacs et al., "Outdoors Augmented Reality on Mobile Phone using Loxel-Based Visual Feature Organization," 2008, ACM, 8 pages. cited by applicant

Livingston et al., "An Augmented Reality System for Military Operations in Urban Terrain," 2002, Proceedings of the Interservice/Industry Training, Simulation, & Education Conference , 8 pages. cited by applicant

Sappa et al., "Chapter 3—Stereo Vision Camera Pose Estimation for On-Board Applications," 2007, I-Tech Education and Publishing, 12 pages. cited by applicant

Light et al., "Chutney and Relish: Designing to Augment the Experience of Shopping at a Farmers' Market," 2010, ACM, 9 pages. cited by applicant

Bell et al., "View Management for Virtual and Augmented Reality," 2001, ACM, 11 pages. cited by applicant

Cyganek et al., "An Introduction to 3D Computer Vision Techniques and Algorithms," 2009, John Wiley & Sons, Ltd, 502 pages. cited by applicant

Lu et al., "Foreground and Shadow Occlusion Handling for Outdoor Augmented Reality," 2010, IEEE, 10 pages. cited by applicant

Lecocq-Botte et al., "Chapter 25—Image Processing Techniques for Unsupervised Pattern Classification," 2007, pp. 467-488. cited by applicant

Lombardo, "Hyper-NPSNET: embedded multimedia in a 3D virtual world," 1993, 83 pages. cited by applicant

Doignon, "Chapter 20—An Introduction to Model-Based Pose Estimation and 3-D Tracking Techniques," 2007, IEEE, I-Tech Education and Publishing, 26 pages cited by applicant

Forsyth et al., "Computer Vision a Modern Approach, Second Edition," 2012, Pearson Education, Inc., Prentice Hall, 793 pages. cited by applicant

Breen et al., "Interactive Occlusion and Collision of Real and Virtual Objects in Augmented Reality," 1995, ECRC, 22 pages. cited by applicant

Breen et al., "Interactive Occlusion and Automatic Object Placement for Augmented Reality," 1996, Eurographics (vol. 15, No. 3), 12 pages. cited by applicant

Pratt et al., "Insertion of an Articulated Human into a Networked Virtual Environment," 1994, Proceedings of the 1994 AI, Simulation and Planning in High Autonomy Systems Conference, 12 pages. cited by applicant

Schmalstieg et al., "Bridging Multiple User Interface Dimensions with Augmented Reality," 2000, IEEE, 10 pages. cited by applicant

Brutzman et al., "Virtual Reality Transfer Protocol (VRTP) Design Rationale," 1997, Proceedings of the IEEE Sixth International Workshop on Enabling Technologies, 10 pages. cited by applicant

Brutzman et al., "Internetwork Infrastructure Requirements for Virtual Environments," 1997, National Academy Press, 12 pages. cited by applicant

Han, "Chapter 1—Real-Time Object Segmentation of the Disparity Map Using Projection-Based Region Merging," 2007, I-Tech Education and Publishing, 20 pages. cited by applicant

George et al., "A Computer-Driven Astronomical Telescope Guidance and Control System with Superimposed Star Field and Celestial Coordinate Graphics Display," 1989, J. Roy. Astron. Soc. Can., The Royal Astronomical Society of Canada, 10 pages. cited by applicant

Marder-Eppstein et al., "The Office Marathon: Robust Navigation in an Indoor Office Environment," 2010, IEEE International Conference on Robotics and Automation, 8 pages. cited by applicant

Barba et al., "Lessons from a Class on Handheld Augmented Reality Game Design," 2009, ACM, 9 pages. cited by applicant

Reitmayr et al., "Collaborative Augmented Reality for Outdoor Navigation and Information Browsing," 2004, 12 pages. cited by applicant

Reitmayr et al., "Going out: Robust Model-based Tracking for Outdoor Augmented Reality," 2006, IEEE, 11 pages. cited by applicant

Regenbrecht et al., "Interaction in a Collaborative Augmented Reality Environment," 2002, CHI, 2 pages. cited by applicant

Herbst et al., "TimeWarp: Interactive Time Travel with a Mobile Mixed Reality Game," 2008, ACM, 11 pages. cited by applicant

Loomis et al., "Personal Guidance System for the Visually Impaired using GPS, GIS, and VR Technologies," 1993, VR Conference Proceedings , 8 pages. cited by applicant

Rekimoto, "Transvision: A hand-held augmented reality system for collaborative design," 1996, Research Gate, 7 pages. cited by applicant

Morse et al., "Multicast Grouping for Data Distribution Management," 2000, Proceedings of the Computer Simulation Methods and Applications Conference, 7 pages. cited by applicant

Morse et al., "Online Multicast Grouping for Dynamic Data Distribution Management," 2000, Proceedings of the Fall

2000 Simulation Interoperability Workshop, 11 pages. cited by applicant

Cheverst et al., “Developing a Context-aware Electronic Tourist Guide: Some Issues and Experiences,” 2000, Proceedings of the Fall 2000 Simulation Interoperability Workshop, 9 pages. cited by applicant

Squire et al., “Mad City Mystery: Developing Scientific Argumentation Skills with a Place-based Augmented Reality Game on Handheld Computers,” 2007, Springer, 25 pages. cited by applicant

Romero et al., “Chapter 10—A Tutorial on Parametric Image Registration,” 2007, I-Tech Education and Publishing, 18 pages. cited by applicant

Rosenberg, “The Use of Virtual Fixtures as Perceptual Overlays to Enhance Operator Performance in Remote Environments,” 1992, Air Force Material Command, 53 pages. cited by applicant

Livingston et al., “Mobile Augmented Reality: Applications and Human Factors Evaluations,” 2006, Naval Research Laboratory, 32 pages. cited by applicant

Gabbard et al., “Resolving Multiple Occluded Layers in Augmented Reality,” 2003, IEEE, 11 pages. cited by applicant

Wloka et al., “Resolving Occlusion in Augmented Reality,” 1995, ACM, 7 pages. cited by applicant

Zyda, “Vrais Panel on Networked Virtual Environments,” Proceedings of the 1995 IEEE Virtual Reality Annual Symposium, 2 pages, 1995. cited by applicant

Zyda et al., “NPSNET-HUMAN: Inserting the Human into the Networked Synthetic Environment,” 1995, Proceedings of the 13th DIS Workshop, 5 pages. cited by applicant

International Search Report and Written Opinion issued in International Application No. PCT/US2013/034164 dated Aug. 27, 2013, 11 pages. cited by applicant

Office Action issued in Japanese Application No. 2014-558993 dated Sep. 24, 2015, 7 pages. cited by applicant

Office Action issued in Japanese Application No. 2014-542591 dated Feb. 23, 2016, 8 pages. cited by applicant

Office Action issued in Japanese Application No. 2014-542591 dated Jul. 7, 2015, 6 pages. cited by applicant

Supplementary European Search Report issued in European Application No. 13854232.9 dated Jul. 24, 2015, 8 pages. cited by applicant

Supplementary European Search Report issued in European Application No. 12852089.7 dated Mar. 13, 2015, 8 pages. cited by applicant

Zhu et al., “Design of the Promo Pad: an Automated Augmented Reality Shopping Assistant,” 12th Americas Conference on Information Systems, Aug. 4-6, 2006, 16 pages. cited by applicant

International Search Report and Written Opinion issued in International Application No. PCT/US2012/066300 dated Feb. 19, 2013, 9 pages. cited by applicant

International Preliminary Report on Patentability issued in International Application No. PCT/US2012/066300 dated Feb. 19, 2014, 12 pages. cited by applicant

Hardawar, “Naratte's Zoosh enables NFC with just a speaker and microphone,” Venture Beat News, <https://venturebeat.com/2011/06/19/narattes-zoosh-enables-nfc-with-just-a-speaker-and-microphone/>, 24 pages, 2011. cited by applicant

Monahan, “Apple iPhone EasyPay Mobile Payment Rollout May Delay NFC,” Javelin Strategy & Research Blog, Nov. 15, 2011, 3 pages. cited by applicant

“Augmented GeoTravel—Support,” <https://web.archive.org/web/20110118072624/http://www.augmentedworks.com/en/augmented-geotravel/augmented-geotravel-support>, 2 pages, 2010. cited by applicant

“Augmented GeoTravel,” https://web.archive.org/web/20200924232145/https://en.wikipedia.org/wiki/Augmented_GeoTravel, 2 pages, 2020. cited by applicant

“AugmentedWorks—iPhone Apps Travel Guide with AR: Augmented Geo Travel 3.0.0!,” <https://web.archive.org/web/20110128180606/http://www.augmentedworks.com/>, 3 pages, 2010. cited by applicant

Rockwell et al., “Campus Mysteries: Serious Walking Around,” Loading . . . The Journal of the Canadian Game Studies Association, 2013, vol. 7, No. 12, 18 pages. cited by applicant

Honkamaa et al., “A Lightweight Approach for Augmented Reality on Camera Phones using 2D Images to Simulate 3D,” Proceedings of the 6th international conference on Mobile and ubiquitous multimedia, 2007, pp. 155-159. cited by applicant

Höllerer et al., “Mobile Augmented Reality,” Telegeoinformatics: Location-based computing and services, vol. 21, 2004, 39 pages. cited by applicant

Raskar et al., “The Office of the Future: A Unified Approach to Image-Based Modeling and Spatially Immersive Displays,” Proceedings of the 25th annual conference on Computer graphics and interactive techniques, 1998, 10 pages. cited by applicant

Loomis et al., “Personal Guidance System for the Visually Impaired using GPS, GIS, and VR Technologies,” Proceedings of the first annual ACM conference on Assistive technologies, 1994, 5 pages. cited by applicant

Feiner et al., “A Touring Machine: Prototyping 3D Mobile Augmented Reality Systems for Exploring the Urban Environment,” *Personal Technologies*, vol. 1, 1997, 8 pages. cited by applicant

Screen captures from YouTube video clip entitled “LiveSight for Here Maps—Demo on Nokia Lumia 928,” 1 page, uploaded on May 21, 2013 by user “Mark Guim”. Retrieved from Internet: <<https://www.youtube.com/watch?v=Wf59vblvGmA>>. cited by applicant

Screen captures from YouTube video clip entitled “Parallel Kingdom Cartography Sneak Peek,” 1 page, uploaded on Oct. 29, 2010 by user “PerBlueInc”. Retrieved from Internet: <<https://www.youtube.com/watch?v=LORdGh4aYis>>. cited by applicant

Screen captures from YouTube video clip entitled “Parallel Kingdom—Video 4—Starting Your Territory.mp4,” 1 page, uploaded on Aug. 24, 2010 by user “PerBlueInc”. Retrieved from Internet: <<https://www.youtube.com/watch?app=desktop&v=5zPXKo6yFzM>>. cited by applicant

Screen captures from YouTube video clip entitled “Parallel Kingdom—Video 8—Basics of Trading.mp4,” 1 page, uploaded on Aug. 24, 2010 by user “PerBlueInc”. Retrieved from Internet: <<https://www.youtube.com/watch?v=z6YcmMZvHbl>>. cited by applicant

Screen captures from YouTube video clip entitled “Parallel Tracking and Mapping for Small AR Workspaces (PTAM)—extra,” 1 page, uploaded on Nov. 28, 2007 by user “ActiveVision Oxford”. Retrieved from Internet: <<https://www.youtube.com/watch?v=Y9HMn6bd-v8>>. cited by applicant

Screen captures from Vimeo video clip entitled “Tabletop Speed Trailer,” 1 page, uploaded on Jun. 5, 2013 by user “Dekko”. Retrieved from Internet: <<https://vimeo.com/67737843>>. cited by applicant

Screen captures from YouTube video clip entitled “Delorme PN-40: Viewing maps and Imagery,” 1 page, uploaded on Jan. 21, 2011 by user “Take a Hike GPS”. Retrieved from Internet: <<https://www.youtube.com/watch?v=cMoKKfGDw4s>>. cited by applicant

Screen captures from YouTube video clip entitled “Delorme Earthmate PN-40: Creating Waypoints,” 1 page, uploaded on Nov. 22, 2010 by user “Take a Hike GPS”. Retrieved from Internet: <<https://www.youtube.com/watch?v=rGz-nFdaO9Y>>. cited by applicant

Screen captures from YouTube video clip entitled “Google Maps Navigation (Beta),” 1 page, uploaded on Oct. 27, 2009 by user “Google”. Retrieved from Internet: <https://www.youtube.com/watch?v=tG XK4jKN_jY>. cited by applicant

Screen captures from YouTube video clip entitled “Google Maps for mobile Layers,” 1 page, uploaded on Oct. 5, 2009 by user “Google”. Retrieved from Internet: <<https://www.youtube.com/watch?v=1W90u0Y1HGI>>. cited by applicant

Screen captures from YouTube video clip entitled “Introduction of Sekai Camera,” 1 page, uploaded on Nov. 7, 2010 by user “tonchidot”. Retrieved from Internet: <<https://www.youtube.com/watch?v=oxnKOQkWwF8>>. cited by applicant

Screen captures from YouTube video clip entitled “Sekai Camera for iPad,” 1 page, uploaded on Aug. 17, 2010 by user “tonchidot”. Retrieved from Internet: <<https://www.youtube.com/watch?v=YGwyhEK8mV8>>. cited by applicant

Screen captures from YouTube video clip entitled “TechCrunch 50 Presentation “SekaiCamera” by TonchiDot,” 1 page, uploaded on Oct. 18, 2008 by user “tonchidot”. Retrieved from Internet: <<https://www.youtube.com/watch?v=FKgJTJojVEw>>. cited by applicant

Screen captures from YouTube video clip entitled “Ville Vesterinen—Shadow Cities,” 1 page, uploaded on Feb. 4, 2011 by user “momoams”. Retrieved from Internet: <<https://www.youtube.com/watch?v=QJ1BsgoKYew>>. cited by applicant

Screen captures from YouTube video clip entitled ““Subway”: Star Wars Arcade: Falcon Gunner Trailer #1,” 1 page, uploaded on Nov. 3, 2010 by user “Im/nl Studios”. Retrieved from Internet: <<https://www.youtube.com/watch?v=CFSMXk8Dw10>>. cited by applicant

Screen captures from YouTube video clip entitled “Star Wars Augmented Reality: TIE Fighters Attack NYC!,” 1 page, uploaded on Nov. 3, 2010 by user “Im/nl Studios”. Retrieved from Internet: <<https://www.youtube.com/watch?v=LoodrUC05r0>>. cited by applicant

Screen captures from YouTube video clip entitled “Streetmuseum,” 1 page, uploaded on Dec. 1, 2010 by user “Jack Kerruish”. Retrieved from Internet: <<https://www.youtube.com/watch?v=qSfATEZiUYo>>. cited by applicant

Screen captures from YouTube video clip entitled “UFO on Tape iPhone Gameplay Review—AppSpy.com,” 1 page, uploaded on Oct. 5, 2010 by user “Pocket Gamer”. Retrieved from Internet: <<https://www.youtube.com/watch?v=Zv4J3ucwyJg>>. cited by applicant

U.S. Appl. No. 18/378,977, filed Oct. 11, 2023. cited by applicant

U.S. Appl. No. 18/385,889, filed Oct. 31, 2023. cited by applicant

Macedonia et al., “A Taxonomy for Networked Virtual Environments,” 1997, *IEEE Multimedia*, 20 pages. cited by applicant

Macedonia, “A Network Software Architecture for Large Scale Virtual Environments,” 1995, 31 pages. cited by applicant

Macedonia et al., “NPSNET: A Network Software Architecture for Large Scale Virtual Environments,” 1994, *Proceedings of the 19th Army Science Conference*, 24 pages. cited by applicant

Macedonia et al., "NPSNET: A Multi-Player 3D Virtual Environment Over the Internet," 1995, ACM, 3 pages. cited by applicant

Macedonia et al., "Exploiting Reality with Multicast Groups: A Network Architecture for Large-scale Virtual Environments," Proceedings of the 1995 IEEE Virtual Reality Annual Symposium, 14 pages, 1995. cited by applicant

Paterson et al., "Design, Implementation and Evaluation of Audio for a Location Aware Augmented Reality Game," 2010, ACM, 9 pages. cited by applicant

Organisciak et al., "Pico Safari: Active Gaming in Integrated Environments," Jul. 19, 2016, 22 pages. cited by applicant

Raskar et al., "Spatially Augmented Reality," 1998, 8 pages. cited by applicant

Raskar et al., "The Office of the Future: A Unified Approach to Image-Based Modeling and Spatially Immersive Displays," 1998, Computer Graphics Proceedings, Annual Conference Series, 10 pages. cited by applicant

Grasset et al., "MARE: Multiuser Augmented Reality Environment on table setup," 2002, 2 pages. cited by applicant

Behringer et al., "A Wearable Augmented Reality Testbed for Navigation and Control, Built Solely with Commercial-Off-The-Shelf (COTS) Hardware," International Symposium in Augmented Reality (ISAR 2000) in München (Munich), Oct. 5-6, 2000, 9 pages. cited by applicant

Behringer et al., "Two Wearable Testbeds for Augmented Reality: itWARNS and WIMMIS," International Symposium on Wearable Computing (ISWC 2000), Atlanta, Oct. 16-17, 2000, 3 pages. cited by applicant

Hartley et al., "Multiple View Geometry in Computer Vision, Second Edition," Cambridge University Press , 673 pages, 2004. cited by applicant

Wetzel et al., "Guidelines for Designing Augmented Reality Games," 2008, ACM, 9 pages. cited by applicant

Kasahara et al., "Second Surface: Multi-user Spatial Collaboration System based on Augmented Reality," 2012, Research Gate, 5 pages. cited by applicant

Diverdi et al., "Envisor: Online Environment Map Construction for Mixed Reality," 8 pages, 2008. cited by applicant

Benford et al., "Understanding and Constructing Shared Spaces with Mixed-Reality Boundaries," 1998, ACM Transactions on Computer-Human Interaction, vol. 5, No. 3, 40 pages. cited by applicant

Mann, "Humanistic Computing: "WearComp" as a New Framework and Application for Intelligent Signal Processing," 1998, IEEE, 29 pages. cited by applicant

Feiner et al., "Knowledge-Based Augmented Reality," 1993, Communications of the ACM , 68 pages. cited by applicant

Bible et al., "Using Spread-Spectrum Ranging Techniques for Position Tracking in a Virtual Environment," 1995, Proceedings of Network Realities, 16 pages. cited by applicant

Starner et al., "Mind-Warping: Towards Creating a Compelling Collaborative Augmented Reality Game," 2000, ACM, 4 pages. cited by applicant

Höllerer et al., "Chapter Nine—Mobile Augmented Reality," 2004, Taylor & Francis Books Ltd. , 39 pages. cited by applicant

Langlotz et al., "Online Creation of Panoramic Augmented-Reality Annotations on Mobile Phones," 2012, IEEE, 9 pages. cited by applicant

Kuroda et al., "Shared Augmented Reality for Remote Work Support," 2000, IFAC Manufacturing, 5 pages. cited by applicant

Ramirez et al., "Chapter 5—Soft Computing Applications in Robotic Vision Systems," 2007, I-Tech Education and Publishing, 27 pages. cited by applicant

Lepetit et al., "Handling Occlusion in Augmented Reality Systems: A Semi-Automatic Method," 2000, IEEE, 11 pages. cited by applicant

Zhu et al., "Personalized In-store E-Commerce with the PromoPad: an Augmented Reality Shopping Assistant," 2004, Electronic Journal for E-commerce Tools, 20 pages. cited by applicant

Broll et al., "Toward Next-Gen Mobile AR Games," 2008, IEEE, 10 pages. cited by applicant

Lee et al., "Exploiting Context-awareness in Augmented Reality Applications," International Symposium on Ubiquitous Virtual Reality, 4 pages, 2008. cited by applicant

Tian et al., "Real-Time Occlusion Handling in Augmented Reality Based on an Object Tracking Approach," www.mdpi.com/journal/sensors, 16 pages, 2010. cited by applicant

Szalavári et al., "Collaborative Gaming in Augmented Reality," 1998, ACM, 20 pages. cited by applicant

Sheng et al., "A Spatially Augmented Reality Sketching Interface for Architectural Daylighting Design," 2011, IEEE, 13 pages. cited by applicant

Szeliski et al., "Computer Vision: Algorithms and Applications," 2010, Springer, 874 pages. cited by applicant

Avery et al., "Improving Spatial Perception for Augmented Reality X-Ray Vision," 2009, IEEE, 4 pages. cited by applicant

Brutzman et al., "Intemetwork Infrastructure Requirements for Virtual Environments," 1995, Proceedings of the Virtual Reality Modeling Language (VRML) Symposium, 11 pages. cited by applicant

Selman, "Java 3D Programming," 2002, Manning Publications, 352 pages. cited by applicant

Bradski et al., "Learning OpenCV," 2008, O'Reilly Media, 572 pages. cited by applicant

Schmeil et al., "MARA—Mobile Augmented Reality-Based Virtual Assistant," 2007, IEEE Virtual Reality Conference 2007, 5 pages. cited by applicant

Macedonia et al., "NPSNET: A Network Software Architecture for Large Scale Virtual Environments," 1994, Presence, Massachusetts Institute of Technology, 30 pages. cited by applicant

Guan et al., "Spherical Image processing for Immersive Visualisation and View Generation," 2011, Thesis submitted to the University of Lancashire, 133 pages. cited by applicant

Sowizral et al., "The Java 3D API Specification—Second Edition," 2000, Sun Microsystems, Inc., 664 pages. cited by applicant

Moore, "A Tangible Augmented Reality Interface to Tiled Street Maps and its Usability Testing," 2006, Springer-Verlag, 18 pages. cited by applicant

Ismail et al., "Multi-user Interaction in Collaborative Augmented Reality for Urban Simulation," 2009, IEEE Computer Society, 10 pages. cited by applicant

Organisciak et al., "Pico Safari—Active Garning in Integrated Environments," 2011, SDH-SEMI (available at <https://www.slideshare.net/PeterOrganisciakpico-safari-sdsemi-2011>). cited by applicant

Davison, "Chapter 7 Walking Around the Models," Pro Java™ 6 3D Game Development Java 3D, 23 pages, 2007. cited by applicant

"Archive for the 'Layers' Category," May 29, 2019, Layar. cited by applicant

Neider et al., "The Official Guide to Learning OpenGL, Version 1.1," Addison Wesley Publishing Company, 616 pages, 1997. cited by applicant

Singhal et al., "Networked Virtual Environments—Design and Implementation", ACM PRESS, Addison Wesley, 1999, 368 pages. cited by applicant

Neider et al., "OpenGL programming guide," 1993, vol. 478, 438 pages. cited by applicant

Julier et al., "BARS: Battlefield Augmented Reality System," Advanced Information Technology (Code 5580), Naval Research Laboratory, 2000, 7 pages. cited by applicant

Baillet et al., "Authoring of Physical Models Using Mobile Computers," Naval Research Laboratory, 2001, IEEE, 8 Pages. cited by applicant

Cheok et al., "Human Pacman: a mobile, wide-area entertainment system based on physical, social, and ubiquitous computing," Springer-Verlag London Limited 2004, 11 pages. cited by applicant

Davidson, "Pro Java™ 6 3D Game Development Java 3D Jogl, Jinput, and JOAL APIs," APRESS, 508 pages, 2007. cited by applicant

Boger, "Are Existing Head-Mounted Displays 'Good Enough'?", Sensics, Inc., 2007, 11 pages. cited by applicant

Boger, "The 2008 HMD Survey: Are We There Yet?" Sensics, Inc., 2008, 14 pages. cited by applicant

Boger, "Cutting the Cord: the 2010 Survey on using Wireless Video with Head-Mounted Displays," Sensics, Inc., 2008, 10 pages. cited by applicant

Bateman, "The Essential Guide to 3D in Flash," 2010, Friends of Ed—an Apress Company, 275 pages. cited by applicant

Guan, "Spherical Image Processing for Immersive Visualisation and View Generation," Thesis Submitted to the University of Central Lancashire, 133 pages, 2011. cited by applicant

Magerkurth, "Proceedings of PerGames—Second International Workshop on Gaming Applications in Pervasive Computing Environments," www.pergames.de., 2005, 119 pages. cited by applicant

Avery, "Outdoor Augmented Reality Gaming on Five Dollars a Day," www.pergames.de., 2005, 10 pages. cited by applicant

Ivanov, "Away 3D 3.6 Cookbook," 2011, Pakt Publishing, 480 pages. cited by applicant

Azuma et al., "Recent Advances in Augmented Reality," IEEE Computer Graphics and Applications, 14 pages, 2001. cited by applicant

Azuma, "A Survey of Augmented Reality," Presence: Teleoperators and Virtual Environments, 48 pages, 1997. cited by applicant

Azuma, "The Challenge of Making Augmented Reality Work Outdoors," In Mixed Reality: Merging Real and Virtual Worlds. Yuichi Ohta and Hideyuki Tamura (ed.), Springer-Verlag, 1999. Chp 21 pp. 379-390, 10 pages. cited by applicant

Bell et al., "Interweaving Mobile Games With Everyday Life," Proc. ACM CHI, 2006, 10 pages. cited by applicant

Bonamico, "A Java-based MPEG-4 Facial Animation Player," Proc Int Conf Augmented Virtual Reality& 3D Imaging, 4 pages, 2001. cited by applicant

Broll, "Meeting Technology Challenges of Pervasive Augmented Reality Games," ACM, 2006, 13 pages. cited by applicant

Brooks, "What's Real About Virtual Reality?," IEEE, Nov./Dec. 1999, 12 pages. cited by applicant

Julier et al., "The Need for AI: Intuitive User Interfaces for Mobile Augmented Reality Systems," 2001, ITT Advanced

Engineering Systems, 5 pages. cited by applicant

Burdea et al., "Virtual Reality Technology: Second Edition," 2003, John Wiley & Sons, Inc., 134 pages. cited by applicant

Butterworth et al., "3DM: A Three Dimensional Modeler Using a Head-Mounted Display," ACM, 1992, 5 pages. cited by applicant

Lee et al., "CAMAR 2.0: Future Direction of Context-Aware Mobile Augmented Reality," 2009, IEEE, 5 pages. cited by applicant

Cheok et al., "Human Pacman: A Mobile Entertainment System with Ubiquitous Computing and Tangible Interaction over a Wide Outdoor Area," 2003, Springer-Verlag, 16 pages. cited by applicant

Hezel et al., "Head Mounted Displays for Virtual Reality," Feb. 1993, MITRE, 5 pages. cited by applicant

McQuaid, "Everquest Shadows of Luclin Game Manual," 2001, Sony Computer Entertainment America, Inc., 15 pages, 2001. cited by applicant

"Everquest Trilogy Manual," 2001, Sony Computer Entertainment America, Inc., 65 pages. cited by applicant

Kellner et al., "Geometric Calibration of Head-Mounted Displays and its Effects on Distance Estimation," Apr. 2012, IEEE Transactions on Visualization and Computer Graphics, vol. 18, No. 4, IEEE Computer Society, 8 pages. cited by applicant

L. Gutierrez et al., "Far-Play: a framework to develop Augmented/Alternate Reality Games," Second IEEE Workshop on Pervasive Collaboration and Social Networking, 2011, 6 pages. cited by applicant

Feiner et al., "A Touring Machine: Prototyping 3D Mobile Augmented Reality Systems for Exploring the Urban Environment," InProc ISWC '97 (Int. Symp. on Wearable Computing), Cambridge, MA, Oct. 13-14, 1997, pp. 74-81, 8 pages. cited by applicant

Fisher et al., "Virtual Environment Display System," Oct. 23-24, 1986, ACM, 12 pages. cited by applicant

Fuchs et al., "Virtual Reality: Concepts and Technologies," 2011, CRC Press, 132 pages. cited by applicant

Gabbard et al., "Usability Engineering: Domain Analysis Activities for Augmented Reality Systems," 2002, The Engineering Reality of Virtual Reality, Proceedings SPIE Volume 4660, Stereoscopic Displays and Virtual Reality Systems IX, 13 pages. cited by applicant

Gledhill et al., "Panoramic imaging—a review," 2003, Elsevier Science Ltd., 11 pages. cited by applicant

Gotow et al., "Addressing Challenges with Augmented Reality Applications on Smartphones," Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, 2010, 14 pages. cited by applicant

"GPS accuracy and Layar usability testing," 2010, mediaLABamsterdam, 7 pages. cited by applicant

Gradecki, "The Virtual Reality Construction Kit," 1994, Wiley & Sons Inc, 100 pages. cited by applicant

Heymann et al., "Representation, Coding and Interactive Rendering of High-Resolution Panoramic Images and Video Using MPEG-4," 2005, 5 pages. cited by applicant

Hollands, "The Virtual Reality Homebrewer's Handbook," 1996, John Wiley & Sons, 213 pages. cited by applicant

Hollerer et al., "User Interface Management Techniques for Collaborative Mobile Augmented Reality," Computers and Graphics 25(5), Elsevier Science Ltd, Oct. 2001, pp. 799-810, 9 pages. cited by applicant

Holloway et al., "Virtual Environments: A Survey of the Technology," Sep. 1993, 59 pages. cited by applicant

Strickland, "How Virtual Reality Gear Works," Jun. 7, 2009, How Stuff Works, Inc., 3 pages. cited by applicant

Hurst et al., "Mobile 3D Graphics and Virtual Reality Interaction," 2011, ACM, 8 pages. cited by applicant

"Human Pacman-Wired NextFest," 2005, Wired. cited by applicant

Cheok et al., "Human Pacman: A Sensing-based Mobile Entertainment System with Ubiquitous Computing and Tangible Interaction," 2000, ACM, 12 pages. cited by applicant

Basu et al., "Immersive Virtual Reality On-The-Go," 2013, IEEE Virtual Reality, 2 pages. cited by applicant

"Inside QuickTime—The QuickTime Technical Reference Library—QuickTime VR," 2002, Apple Computer Inc., 272 pages. cited by applicant

Bayer et al., "Chapter 3: Introduction to Helmet-Mounted Displays," 62 pages, 2009. cited by applicant

Iovine, "Step Into Virtual Reality", 1995, Windcrest/McGraw-Hill, 106 pages. cited by applicant

Jacobson et al., "Garage Virtual Reality," 1994, Sams Publishing, 134 pages. cited by applicant

Lin, "How is Nike+ Heat Map Calculated?," howtonike.blogspot.com/2012/06/how-is-nike-heat-map-calculated.html, 4 pages, 2012. cited by applicant

"Map your run with new Nike+ GPS App," Nike News, Sep. 7, 2010, 3 pages. cited by applicant

Savov, "App review: Nike+ GPS," <https://www.engadget.com/2010-09-07-app-review-nike-gps.html>, 4 pages, 2010. cited by applicant

Lutz, "Nokia reveals new City Lens augmented reality app for Windows Phone 8 lineup," <https://www.engadget.com/2012-09-11-nokia-reveals-new-city-lens-for-windows-phone-8.html>, 3 pages, 2012. cited by applicant

Nayan, "Bytes: Livesight update integrates "City Lens" to Here Maps !! Nokia announces partnership with "Man of

steel“, releases promo,” <https://nokiapoweruser.com/bytes-livesight-update-integrates-city-lens-to-here-maps-nokia-announces-partnership-with-man-of-steel-release-promo-video/>, 4 pages, 2013. cited by applicant

Webster, “Nokia's City Lens augmented reality app for Lumia Windows Phones comes out of beta,” <https://www.theverge.com/2012/9/2/3287420/nokias-city-lens-ar-app-launch>, 2 pages, 2012. cited by applicant

“Nokia Image Space on video,” <https://blogs.windows.com/devices/2008/09/24/nokia-image-space-on-video/>, 4 pages, 2008. cited by applicant

Montola et al., “Applying Game Achievement Systems to Enhance User Experience in a Photo Sharing Service,” *Proceedings of the 13th International MindTrek Conference: Everyday Life in the Ubiquitous Era*, 2009, pp. 94-97. cited by applicant

Arghire, “Nokia Image Space Now Available for Download,” <https://news.softpedia.com/news/Nokia-Image-Space-Now-Available-for-Download-130523.shtml>, 2 pages, 2009. cited by applicant

Then, “Nokia Image Space adds Augmented Reality for S60,” <https://www.slashgear.com/nokia-image-space-adds-augmented-reality-for-s60-3067185/>, 6 pages, 2009. cited by applicant

Uusitalo et al., “A Solution for Navigating User-Generated Content,” 2009 8th IEEE International Symposium on Mixed and Augmented Reality, 2009, pp. 219-220. cited by applicant

Bhushan, “Nokia Rolls out Livesight to Here Maps,” <https://www.digit.in/news/apps/nokia-rolls-out-livesight-to-here-maps-14740.html>, 2 pages, 2013. cited by applicant

Blandford, “Here Maps adds LiveSight integration to let you “see” your destination,” http://allaboutwindowsphone.com/news/item/17563_HERE_Maps_adds_LiveSight_integ.php, 16 pages, 2013. cited by applicant

Bonetti, “Here brings sight recognition to Maps,” <https://web.archive.org/web/20130608025413/http://conversations.nokia.com/2013/05/21/here-brings-sight-recognition-to-maps/>, 5 pages, 2013. cited by applicant

Burns, “Nokia City Lens released from Beta for Lumia devices,” <https://www.slashgear.com/nokia-city-lens-released-from-beta-for-lumia-devices-1%20246841/>, 9 pages, 2012. cited by applicant

Viswav, “Nokia Details the New LiveSight Experience on Here Maps,” <https://mspoweruser.com/nokia-details-the-new-livesight-experience-on-here-maps/>, 19 pages, 2013. cited by applicant

Viswav, “Nokia Announces LiveSight, An Augmented Reality Technology,” <https://mspoweruser.com/nokia-announces-livesight-an-augmented-reality-technology/>, 19 pages, 2012. cited by applicant

Varma, “Nokia Here Map gets integrated with LiveSight Augmented Reality feature,” <https://www.datareign.com/nokia-here-map-integrate-livesight-augmented-reality-feature/>, 5 pages, 2013. cited by applicant

Bosma, “Nokia works on mobile Augmented Reality (AR),” <https://www.extendlimits.nl/en/article/nokia-works-on-mobile-augmented-reality-ar>, 6 pages, 2006. cited by applicant

Greene, “Hyperlinking Reality via Phones,” <https://www.technologyreview.com/2006/11/20/273250/hyperlinking-reality-via-phones/>, 11 pages, 2006. cited by applicant

Knight, “Mapping the world on your phone,” <https://www.cnn.com/2007/TECH/science/05/23/Virtualmobile.1/>, 2 pages 2007. cited by applicant

“MARA,” <https://web.archive.org/web/20100531083640/http://www.research.nokia.com:80/research/projects/mara>, 3 pages, 2010. cited by applicant

Maubon, “A little bit of history from 2006: Nokia MARA project,” <https://www.augmented-reality.fr/2009/03/un-petit-peu-dhistoire-de-2006-projet-mara-de-nokia/>, 7 pages, 2009. cited by applicant

“Nokia's MARA Connects The Physical World Via Mobile,” <https://theponderingprimate.blogspot.com/2006/11/nokias-mara-connects-physical-world.html>, 14 pages, 2006. cited by applicant

Patro et al., “The anatomy of a large mobile massively multiplayer online game,” *Proceedings of the first ACM international workshop on Mobile gaming*, 2012, 6 pages. cited by applicant

Schumann et al., “Mobile Gaming Communities: State of the Art Analysis and Business Implications,” *Central European Conference on Information and Intelligent Systems*, 2011, 8 pages. cited by applicant

Organisciak, “Pico Safari: Active Gaming in Integrated Environments,” <https://organisciak.wordpress.com/2016/07/19/pico-safari-active-gaming-in-integrated-environments/>, 21 pages, 2016. cited by applicant

“Plundr,” <https://web.archive.org/web/20110110032105/areacodeinc.com/projects/plundr/>, 3 pages, 2007. cited by applicant

Caoili et al., “Plundr: Dangerous Shores' location-based gaming weighs anchor on the Nintendo DS,” <https://www.engadget.com/2007-06-03-plundr-dangerous-shores-location-based-gaming-weighs-anchor-on-the-nintendo-ds.html>, 2 pages, 2007. cited by applicant

Miller, “Plundr, first location-based DS game, debuts at Where 2.0,” <https://www.engadget.com/2007-06-04-plundr-first-location-based-ds-game-debuts-at-where-2-0.html>, 4 pages, 2007. cited by applicant

Blösch et al., “Vision Based MAV Navigation in Unknown and Unstructured Environments,” 2010 IEEE International

Conference on Robotics and Automation, 2010, 9 pages. cited by applicant

Castle et al., “Video-rate Localization in Multiple Maps for Wearable Augmented Reality,” 2008 12th IEEE International Symposium on Wearable Computers, 8 pages, 2008. cited by applicant

Klein et al. “Parallel Tracking and Mapping for Small AR Workspaces,” 2007 6th IEEE and ACM international symposium on mixed and augmented reality, 2007, 10 pages. cited by applicant

Klein et al. “Parallel tracking and mapping on a camera phone,” 2009 8th IEEE International Symposium on Mixed and Augmented Reality, 2009, 4 pages. cited by applicant

Van Den Hengel et al., “In Situ Image-based Modeling,” 2009 8th IEEE International Symposium on Mixed and Augmented Reality, 2009, 4 pages. cited by applicant

Hughes, “Taking social games to the next level,” <https://www.japantimes.co.jp/culture/2010/08/04/general/taking-social-games-to-the-next-level/>, 1 page, 2010. cited by applicant

Kincaid, “TC50 Star Tonchidot Releases Its Augmented Reality Sekai Camera Worldwide,” <https://techcrunch.com/2009/12/21/sekai-camera/>, 9 pages, 2009. cited by applicant

Martin, “Sekai Camera's new reality,” <https://www.japantimes.co.jp/life/2009/10/14/digital/sekai-cameras-new-reality/>, 3 pages, 2009. cited by applicant

Nakamura et al., “Control of Augmented Reality Information Volume by Glabellar Fader,” Proceedings of the 1st Augmented Human international Conference, 2010, 3 pages. cited by applicant

“Anime×Tsutaya×Sekai Camera,” <https://japanesevw.blogspot.com/2010/08/animetsutayasekai-camera.html#links>, 4 pages, 2010. cited by applicant

“AR-RPG(ARPG) ”Sekai Hero“,” <https://japanesevw.blogspot.com/2010/08/ar-rpgarpg-sekai-hero.html#links>, 5 pages, 2010. cited by applicant

Toto, “Augmented Reality App Sekai Camera Goes Multi-Platform. Adds API and Social Gaming,” <https://techcrunch.com/2010/07/14/augmented-reality-app-sekai-camera-goes-multi-platform-adds-api-and-social-gaming/>, 4 pages, 2010. cited by applicant

Hämäläinen, “[Job] Location Based MMORPG server engineers—Grey Area & Shadow Cities,” <https://erlang.org/pipermail/erlang-questions/2010-November/054788.html>, 2 pages, 2010. cited by applicant

Jordan, “Grey Area CEO Ville Vesterinen on building out the success of location-based Finnish hit Shadow Cities,” <https://www.pocketgamer.com/grey-area-news/grey-area-ceo-ville-vesterinen-on-building-out-the-success-of-location-based-fin/>, 4 pages, 2010. cited by applicant

“Shadow Cities,” <https://web.archive.org/web/20101114162700/http://www.shadowcities.com/>, 7 pages, 2010. cited by applicant

Buchanan, “Star Wars: Falcon Gunner iPhone Review,” <https://www.ign.com/articles/2010/11/18/star-wars-falcon-gunner-iphone-review>, 13 pages, 2010. cited by applicant

“THQ Wireless Launches Star Wars Arcade: Falcon Gunner,” https://web.archive.org/web/20101129010405/http://starwars.com/games/videogames/swarcade_falcongunner/index.html, 5 pages, 2010. cited by applicant

Firth, “Play Star Wars over the city: The incredible new game for iPhone that uses camera lens as backdrop for spaceship dogfights,” <https://www.dailymail.co.uk/sciencetech/article-1326564/Star-Wars-Arcade-Falcon-Gunner-iPhone-game-uses-camera-lens-backdrop.html>, 24 pages, 2010. cited by applicant

Grundman, “Star Wars Arcade: Falcon Gunner Review,” <https://www.148apps.com/reviews/star-wars-arcade-falcon-gunner-review/>, 11 pages, 2010. cited by applicant

“Star Wars Arcade: Falcon Gunner,” <https://www.macupdate.com/app/mac/35949/star-wars-arcade-falcon-gunner>, downloaded on Feb. 9, 2021, 5 pages. cited by applicant

Nelson, “THQ Announces ‘Star Wars: Falcon Gunner’ Augmented Reality Shooter,” <https://toucharcade.com/2010/11/04/thq-announces-star-wars-falcon-gunner-augmented-reality-shooter/>, 6 pages, 2010. cited by applicant

Rogers, “Review: Star Wars Arcade: Falcon Gunner,” source.com/2010/12/04/review-star-wars-arcade-falcon-gunner/, downloaded on Feb. 9, 2021, 14 pages, 2010. cited by applicant

Schonfeld, “The First Augmented Reality Star Wars Game, Falcon Gunner, Hits The App Store,” <https://techcrunch.com/2010/11/17/star-wars-iphone-falcon-gunner/>, 15 pages, 2010. cited by applicant

“How it Works,” <https://web.archive.org/web/20130922212452/http://www.strava.com/how-it-works>, 4 pages, 2013. cited by applicant

“Tour,” <https://web.archive.org/web/20110317045223/http://www.strava.com/tour>, 9 pages, 2011. cited by applicant

Eccleston-Brown, “Old London seen with new eyes thanks to mobile apps,” http://news.bbc.co.uk/local/london/hi/things_to_do/newsid_8700000/8700410.stm, 3 pages, 2010. cited by applicant

“Streetmuseum' Museum of London App Offers a New Perspective on the Old,” <https://www.trendhunter.com/trends/streetmuseum-museum-of-london-app>, 6 pages, 2021. cited by applicant

“Museum of London ‘StreetMuseum’ by Brothers and Sisters,” <https://www.campaignlive.co.uk/article/museum-london-streetmuseum-brothers-sisters/1003074>, 11 pages, 2010. cited by applicant

Zhang, “Museum of London Releases Augmented Reality App for Historical Photos,” <https://petapixel.com/2010/05/24/museum-of-london-releases-augmented-reality-app-for-historical-photos/>, 11 pages, 2010. cited by applicant

Lister, “Turf Wars and Fandango among this week’s free iPhone apps,” <https://www.newsreports.com/turf-wars-and-fandango-among-this-week-s-free-iphone-apps/>, 6 pages, 2010. cited by applicant

McCavitt, “Turf Wars iPhone Game Review,” <https://web.archive.org/web/20100227030259/http://www.thegamereviews.com:80/article-1627-Turf-Wars-iPhone-Game-Review.html>, 2 pages, 2010. cited by applicant

“Turf Wars Captures Apple’s iPad,” old.gamegrin.com/game/news/2010/turf-wars-captures-apples-ipad, downloaded on Feb. 5, 2021, 2 pages, 2010. cited by applicant

James, “Turf Wars (iPhone GPS Game) Guide and Walkthrough,” <https://web.archive.org/web/20120114125609/http://gameolosophy.com/games/turf-wars-iphone-gps-game-guide-and-walkthrough>, 3 pages, 2011. cited by applicant

Rachel et al., “Turf Wars’ Nick Baicoianu—Exclusive Interview,” <https://web.archive.org/web/20110101031555/http://www.gamingangels.com/2009/12/turf-wars-nick-baicoianu-exclusive-interview/>, 7 pages, 2009. cited by applicant

Gharriy, “Turf Wars Q&A,” <https://web.archive.org/web/20110822135221/http://blastmagazine.com/the-magazine/gaming/gaming-news/turf-wars-qa/>, 11 pages, 2010. cited by applicant

“Introducing Turf Wars, the Free, GPS based Crime Game for Apple iPhone,” <https://www.ign.com/articles/2009/12/07/introducing-turf-wars-the-free-gps-based-crime-game-for-apple-iphone>, 11 pages, 2009. cited by applicant

“Turf Wars,” <https://web.archive.org/web/20100328171725/http://itunes.apple.com:80/app/turf-wars/id332185049?mt=8>, 3 pages, 2010. cited by applicant

Zungre, “Turf Wars Uses GPS to Control Real World Territory,” <https://web.archive.org/web/20110810235149/http://www.slidetoplay.com/story/turf-wars-uses-gps-to-control-real-world-territory>, 1 page, 2009. cited by applicant

“Turf Wars,” <https://web.archive.org/web/20101220170329/http://turf warsapp.com/>, 1 page, 2010. cited by applicant

“Turf Wars News,” <https://web.archive.org/web/20101204075000/http://turf warsapp.com/news/>, 5 pages, 2010. cited by applicant

“Turf Wars Screenshots,” <https://web.archive.org/web/20101204075000/http://turf warsapp.com/news/>, 5 pages, 2010. cited by applicant

Broida, “UFO on Tape: The game of close encounters,” <https://www.cnet.com/news/ufo-on-tape-the-game-of-close-encounters/>, 7 pages, 2010. cited by applicant

Buchanan, “UFO on Tape Review,” <https://www.ign.com/articles/2010/09/30/ufo-on-tape-review>, 7 pages, 2010. cited by applicant

Nesvadba, “UFO on Tape Review,” <https://www.appspy.com/review/4610/ufo-on-tape>, 3 pages, 2010. cited by applicant

Barry, “Waze Combines Crowdsourced GPS and Pac-Man,” <https://www.wired.com/2010/11/waze-combines-crowdsourced-gps-and-pac-man/>, 2 pages, 2010. cited by applicant

Dempsey, “Waze: Crowdsourcing traffic and roads,” <https://www.gislounge.com/crowdsourcing-traffic-and-roads/>, 10 pages, 2010. cited by applicant

Forrest, “Waze: Make Your Own Maps in Realtime,” <http://radar.oreilly.com/2009/08/waze-make-your-own-maps-in-re.html>, 4 pages, 2009. cited by applicant

Forrest, “Waze: Using groups and gaming to get geodata,” <http://radar.oreilly.com/2010/08/waze-using-groups-and-gaming-t.html>, 3 pages, 2010. cited by applicant

Furchgott, “The Blog; App Warns Drivers of the Mayhem Ahead,” <https://archive.nytimes.com/query.nytimes.com/gst/fullpage-9B07EFDC1E3BF930A25751C0A967908B63.html>, downloaded on Feb. 17, 2021, 2 pages. cited by applicant

Ha, “Driving app Waze turns the highway into a Pac-Man game with ‘Road Goodies’,” <https://venturebeat.com/social/driving-app-waze-turns-the-highway-into-a-pac-man-style-game-with-road-goodies/>, 4 pages, 2009. cited by applicant

Rogers, “Review: Waze for the iPhone,” source.com/2010/08/30/review-waze-for-the-iphone/, downloaded on Feb. 17, 2021, 22 page. cited by applicant

Fox, “What is Wherigo?,” <https://forums.geocaching.com/GC/index.php?/topic/241452-what-is-herigo/>, 4 pages, 2010 cited by applicant

Lenahan, “Create Exciting GPS Adventure Games With Wherigo,” <https://www.makeuseof.com/tag/create-gps->

adventure-games-herigo/?utm_source=twitterfeed&utm_medium=twitter, 15 pages, 2010. cited by applicant

“Developers—Download Wikitude API,”
<https://web.archive.org/web/20110702200814/http://www.wikitude.com/en/developers>, 8 pages, 2010. cited by applicant

Hauser, “Wikitude World Browser,”
<https://web.archive.org/web/20110722165744/http://www.wikitude.com/en/wikitude-world-browser-augmented-reality>, 5 pages, 2010. cited by applicant

Madden, “Professional augmented reality browsers for smartphones: programming for junaio, layar and wikitude,” 2011. 345 pages. cited by applicant

Chen, “Yelp Sneaks Augmented Reality Into iPhone App,” <https://www.wired.com/2009/08/yelp-ar/>, 2 pages, 2009. cited by applicant

Herrman, “Augmented Reality Yelp Will Murder All Other iPhone Restaurant Apps, My Health,”
<https://gizmodo.com/augmented-reality-yelp-will-murder-all-other-iphone-res-5347194>, 5 pages, 2009. cited by applicant

“Easter Egg: Yelp Is the iPhone's First Augmented Reality App,” <https://mashable.com/2009/08/27/yelp-augmented-reality/>, downloaded Feb. 5, 2021, 10 pages. cited by applicant

Metz, “‘Augmented reality’ comes to mobile phones,” <https://www.nbcnews.com/id/wbna33165050>, 10 pages, 2009. cited by applicant

Mortensen, “New Yelp App Has Hidden Augmented Reality Mode,” <https://www.cultofmac.com/15247/new-yelp-app-has-hidden-augmented-reality-mode>, 5 pages, 2009. cited by applicant

Schramm, “Voices that Matter iPhone: How Ben Newhouse created Yelp Monocle, and the future of AR,”
<https://www.engadget.com/2010-04-26-voices-that-matter-iphone-how-ben-newhouse-created-yelp-monocle.html>, 7 pages, 2010. cited by applicant

Hand, “NYC Nearest Subway AR App for iPhone 3GS,” <https://vizworld.com/2009/07/nyc-nearest-subway-ar-app-for-iphone-3gs/>, 7 pages, 2009. cited by applicant

“Acrossair Augmented Reality Browser,” <https://appadvice.com/app/acrossair-augmented-reality/348209004>, 3 pages, 2009. cited by applicant

Schwartz, “Lost in the Subway? Use AcrossAir's Augmented Reality iPhone App,”
https://www.fastcompany.com/1311181/lost-subway-use-acrossairs-augmented-reality-iphone-app?itm_source=parsely-api, 8 pages, 2009. cited by applicant

Hartsock, “Acrossair: Getting There Is Half the Fun,” <https://www.technewsworld.com/story/70502.html>, downloaded on Mar. 12, 2021, 5 pages. cited by applicant

“AugmentedWorks—iPhone Apps Travel Guide with AR: Augmented GeoTravel 3.0.0,”
<https://web.archive.org/web/20110128180606/http://augmentedworks.com/>, 3 pages, 2010. cited by applicant

“Augmented GeoTravel—Features,”
<https://web.archive.org/web/20100909163937/http://www.augmentedworks.com/en/augmented-geotravel/features>, 2 pages, 2010. cited by applicant

Varshney, Upkar. “Location Management for Mobile Commerce Applications in Wireless Internet Environment.” ACM transactions on Internet technology 3.3 (2003): 236-255. Web. (Year: 2003). cited by applicant

Hühn, Arief Ernst et al. “On the Use of Virtual Environments for the Evaluation of Location-Based Applications.”. Proceedings of the SIGCHI Conference on Human Factors in Computing Systems. ACM, 2012. 2569-2578. Web. (Year: 2012). cited by applicant

Jackson, Emily. “Shopper Marketing Techs up; From Geo-Targeting to Augmented Reality Shopping Apps, Brands and Retailers Get Digital.” Strategy (2012): 18-. Print. (Year: 2012). cited by applicant

Warner, Chris. Augmented Reality Helps Retailers Get Personal: Sensors Are First in Line at the Point of Sale. Vol. 56. Advantage Business Media, 2012. Print. (Year: 2012). cited by applicant

“Geko 301,” <http://www.garmin.com/en-US/p/221>, Mar. 16, 2021, 3 pages. cited by applicant

“Introducing myNav: Peter Cooper Village/Stuyvesant Town,” <http://mynav-peter-cooper-villagestuyvesant-town-appstor.io/zh>, Mar. 16, 2021, 2 pages. cited by applicant

“myNav: Peter Cooper Village/Stuyvesant Town,” <https://appadvice.com/app/mynav-peter-cooper-village/383793988>, Mar. 16, 2021, 3 pages. cited by applicant

Harris, “How Does the Nike Plue Wrok?,”
<http://web.archive.org/web/20111104034832/http://www.livesdtron.com80/article533191-how-does-the-nike-plus-work/>, Sep. 2, 2011, 2 pages cited by applicant

“Louvre DNP Museum Lab,”
<http://web.archive.org/web/20130317055849fhttp://www.museumlab.eu/exhibition/10/index.html>, Mar. 17, 2013, 2 pages. cited by applicant

“Wherigo Advanced Concepts,”

Primary Examiner: Tseng; Charles

Attorney, Agent or Firm: Maschoff Brennan

Background/Summary

(1) This application is a continuation of U.S. application Ser. No. 17/587,183, filed Jan. 28, 2022, which is a continuation of U.S. application Ser. No. 16/864,075, filed Apr. 30, 2020, which is a continuation of U.S. application Ser. No. 16/168,419, filed Oct. 23, 2018, which is a continuation of U.S. application Ser. No. 15/794,993, filed Oct. 26, 2017, which is a continuation of U.S. application Ser. No. 15/406,146, filed Jan. 13, 2017, which is a continuation of U.S. application Ser. No. 14/517,728, filed Oct. 17, 2014, which claims priority to U.S. Provisional Application No. 61/892,238, filed Oct. 17, 2013. These and all other extrinsic references referenced herein are incorporated by reference in their entirety.

FIELD OF THE INVENTION

(1) The field of the invention is augmented reality service technologies.

BACKGROUND

(2) The following description includes information that may be useful in understanding the present invention. It is not an admission that any of the information provided herein is prior art or relevant to the presently claimed invention, or that any publication specifically or implicitly referenced is prior art.

(3) As advances in technology continue to be developed, the utilization of Augmented Reality (AR) to enhance experiences is becoming increasingly popular. Various entities have attempted to capitalize on this increasing popularity by providing AR content to users based on specific types of object recognition or location tracking.

(4) For example, U.S. Pat. No. 8,519,844 to Richey et al., filed on Jun. 30, 2010 contemplates accessing first and second location data, wherein the second location data has increased accuracy regarding the location of a device, and communicating augmented data to the device based on the location data.

(5) The '844 patent and all other publications identified herein are incorporated by reference to the same extent as if each individual publication or patent application were specifically and individually indicated to be incorporated by reference. Where a definition or use of a term in an incorporated reference is inconsistent or contrary to the definition of that term provided herein, the definition of that term provided herein applies and the definition of that term in the reference does not apply.

(6) Another example of location based content services, while not directed to AR content, can be found in U.S. Pat. No. 8,321,527 to Martin, et al, filed on Sep. 10, 2009, which describes a system for scheduling content distribution to a mobile device by storing different locations, collecting user location data over a period of time, collecting wireless signal strength data, and scheduling pre-caching of content to the device if the user is predicted to be at a location with poor signal strength.

(7) Still further, various other examples of systems and methods for providing content to a user based on a location or other parameters can be found in International Patent Application Publication Number WO 2013/023705 to Hoffman, et al, filed on Aug. 18, 2011, International Patent Application Publication Number WO 2007/140155 to Leonard, et al, filed on May 21, 2007, U.S. Patent Application Publication Number 2013/0003708 to Ko, et al, filed on Jun. 28, 2011, U.S. Patent Application Publication Number 2013/0073988 to Groten, et al, filed on Jun. 1, 2011, and U.S. Patent Application Publication Number 2013/0124326 to Huang, et al, filed on Nov. 15, 2011.

(8) While some of the known references contemplate refining location identification or pre-caching content based on location information, they fail to consider that areas have various views of interest, and fail to differentiate between sub-areas based on AR content densities. Viewed from another perspective, known location based systems fail to contemplate segmenting an area into clusters based on what is viewable or what AR content is available.

(9) Thus, there is still a need for improved AR service technologies, and especially location based AR service technologies.

SUMMARY OF THE INVENTION

(10) The inventive subject matter provides apparatuses, systems and methods in which AR content is provided to one or more user devices based on at least one of location identification and object recognition. In some

contemplated aspects, the user device could be auto-populated with AR content objects based on a location, and the AR content objects could be instantiated based on object recognition within the location.

(11) One aspect of the inventive subject matter includes a content management system comprising a content management engine coupled with an area database and a content database. The content management engine can be configured to communicate with the databases and perform various steps in order to provide content objects to a device for modification or instantiation.

(12) The area database could be configured to store area data related to an area of interest. This area data could comprise image data, video image data, real-time image data, still image data, signal data (e.g., Compressive Sensing of Signals (CSS) data, Received Signal Strength (RSS), WiFi signal data, beacon signal data, etc.), audio data, an initial map (e.g., CAD drawing, 3-dimensional model, blueprint, etc.), or any other suitable data related to a layout of an area.

(13) The content database could be configured to store augmented reality or other digital content objects of various modalities, including for example, image content objects, video content objects, or audio content objects. It is contemplated that the content objects could be associated with one or more real world objects viewable from an area of interest.

(14) Viewed from another perspective, a content management engine of the inventive subject matter could comprise an AR management engine that is configured to obtain an initial map of an area of interest from the area data within the area database. The step of obtaining the initial map could comprise obtaining a CAD, blueprint, 3-D model, a robot or drone created map, or other representation from the area database itself, or could comprise obtaining area data such as image data, signal data, video data, audio data, views data, viewable object data, points of interest data, field of view data, etc. to generate the initial map.

(15) The AR management engine could then derive a set of views of interest from at least one of the initial map and other area data. The views of interest are preferably representative of where people would, should, or could be looking while navigating through various portions of the area of interest. The views of interest could be derived by the map generation engine, or via recommendations, requests or other inputs of one or more users (e.g., potential viewer, advertiser, manager, developer, etc.), could be created manually by a systems manager or other user, or could be modeled based on some or all of the area data. The views of interest could comprise, among other things, a view-point origin, a field of interest, an owner, metadata, a direction (e.g., a vector, an angle, etc.), an orientation (e.g., pitch, yaw, roll, etc.), a cost, a search attribute, a descriptor set, an object of interest, or any combination or multiples thereof. For example, a view of interest could comprise a view-point origin (i.e., point of view origin), at least one field of interest, and a viewable object of interest. Another view of interest could comprise a view-point origin, at least two fields of interest, and a viewable object of interest.

(16) Once the views of interest have been derived, the AR management engine could obtain a set of AR content objects (e.g., a virtual object, chroma key content, digital image, digital video, audio data, application, script, promotion, advertisement, game, workflow, kinesthetic, tactile, lesson plan, etc.) from the AR content database. Each of the AR content objects will preferably be related to one or more of the derived views of interest. The AR content objects could be selected for obtaining based on one or more of the following: a search query, an assignment of content objects to a view of interest or object of interest within the view, one or more characteristics of the initial map, a context of an intended user of a user (e.g., a potential viewer, advertiser, manager, developer, etc.), or a recommendation, selection or request of a user.

(17) The AR management engine could then establish AR experience clusters within the initial map as a function of the AR content objects obtained and views of interest derived. These clusters will preferably represent a combination of the views of interest and related information, and a density or other characteristic of AR content objects related to the views of interest. Viewed from another perspective, each cluster could represent a subset of the derived views of interest and associated AR content objects.

(18) Based on the AR experience clusters or information related thereto, the AR management engine could generate a tile map comprising tessellated tiles (e.g., regular or non-regular (e.g., semi-regular, aperiodic, etc.), Voronoi tessellation, penrose tessellation, K-means cluster, etc.) that cover at least a portion of the area of interest. Some or all of the tiles could advantageously be individually bound to a subset of the obtained AR content objects, which can comprise overlapping or completely distinct subsets. Additionally or alternatively, the tiles could be associated with one or more of an identification, an owner, an object of interest, a set of descriptors, an advertiser, a cost, or a time. Still further, it is contemplated that the tiles could be dynamic in nature such that the tessellation of the area could change based on an event or a time. Contemplated events include, among other things, a sale, a news event, a publication, a change in inventory, a disaster, a change in advertiser, or any other suitable event. It is also contemplated that a view-point origin, a field of interest, a view or an object of interest could be dynamic in nature.

(19) The AR management engine could further configure a device (e.g., a mobile device, a kiosk, a tablet, a cell

phone, a laptop, a watch, a vehicle, a server, a computer, etc.) to obtain at least a portion of the subset based on the tile map (e.g., based on the device's location in relation to the tiles of a tile map, etc.), and present at least a portion of the AR content objects on a display of the device (e.g., instantiate the object, etc.). It is contemplated that the device could compose a data center and be coupled with a cloud server.

(20) Various objects, features, aspects and advantages of the inventive subject matter will become more apparent from the following detailed description of preferred embodiments, along with the accompanying drawing figures in which like numerals represent like components.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic of a system of the inventive subject matter.
- (2) FIG. 2 is a schematic showing a generation of an initial map of an area of interest.
- (3) FIG. 3 provides an example overview of the derivation of views of interest.
- (4) FIG. 4 is a schematic showing a generation of an AR content database.
- (5) FIG. 5 is a schematic showing a generation of a tessellated area map.
- (6) FIG. 6 is a schematic showing an area tile map based on view of interest clusters.

DETAILED DESCRIPTION

(7) It should be noted that while the following description is drawn to a computer/server based device interaction system, various alternative configurations are also deemed suitable and may employ various computing devices including servers, workstations, clients, peers, interfaces, systems, databases, agents, peers, engines, controllers, modules, or other types of computing devices operating individually or collectively. One should appreciate the use of such terms are deemed to represent computing devices comprising at least one processor configured or programmed to execute software instructions stored on a tangible, non-transitory computer readable storage medium (e.g., hard drive, FPGA, solid state drive, RAM, flash, ROM, memory, distributed memory, etc.). The software instructions preferably configure the computing device to provide the roles, responsibilities, or other functionality as discussed below with respect to the disclosed apparatus. Further, the disclosed technologies can be embodied as a computer program product that includes a non-transitory computer readable medium storing the software instructions that causes a processor to execute the disclosed steps. In especially preferred embodiments, the various servers, systems, databases, or interfaces exchange data using standardized protocols or algorithms, possibly based on HTTP, HTTPS, AES, public-private key exchanges, web service APIs, known financial transaction protocols, or other electronic information exchanging methods. Data exchanges among devices can be conducted over a packet-switched network, the Internet, LAN, WAN, VPN, or other type of packet switched network; a circuit switched network; cell switched network; or other type of network.

(8) One should appreciate that the disclosed techniques provide many advantageous technical effects including providing augmented reality content to a user device based on a precise location of the user device relative to one or more tiles of a tessellated area associated with view(s) of interest.

(9) The following discussion provides many example embodiments of the inventive subject matter. Although each embodiment represents a single combination of inventive elements, the inventive subject matter is considered to include all possible combinations of the disclosed elements. Thus if one embodiment comprises elements A, B, and C, and a second embodiment comprises elements B and D, then the inventive subject matter is also considered to include other remaining combinations of A, B, C, or D, even if not explicitly disclosed.

(10) A system of the inventive subject matter could advantageously identify a location of a device at or near a tile of a tessellated area of interest and auto-populate the device with pre-selected content objects based upon the identified location. Exemplary systems and methods for identifying a location of a user or device within or near a tile can be found in U.S. pre-grant publication number 2014/0011518 Valaee, et al, entitled "System, Method And Computer Program For Dynamic Generation Of A Radio Map" and U.S. pre-grant publication 2012/0149415, to Valaee, et al entitled "System, Method and Computer Program for Anonymous Localization."

(11) Where the device is configured or programmed to capture image or other sensor data (e.g., orientation data, position data, etc.) that indicates that an object is viewable by a user of the device, the system can cause the device to instantiate some or all of the content objects based on an association between the viewable object(s) and the content object(s) (e.g., based on at least one of object recognition, orientation, location, etc.). The instantiated AR content object could be presented in any suitable manner, including for example, as an occlusion mask, behind one or more objects, behind an object and in front of a different object, or as a moving object across an object of interest.

(12) FIG. 1 is a schematic of an exemplary system **100** of the inventive subject matter. System **100** comprises a map generation engine **102**, which can be leveraged by one or more users to capture, generate, or otherwise obtain

area data related to an area of interest. Among other suitable data, area data could comprise image data **112** (e.g., still image data, real-time image data, etc.), video data **114**, signal data **116** (e.g., CSS data, RSS data, WiFi signal data, beacon signal data, etc.), and/or initial maps **118** that could be transmitted to and stored in area database **110** via network **105**. AR management engine **130**, coupled with area database **110** via network **125**, can be configured to obtain an initial map **118A** related to an area of interest from area database **110**, or could be configured to obtain other area data and generate initial map **118A** based on the obtained data.

(13) An area of interest can be considered generally to be a real-world space, area or setting selected within which the processes and functions of the inventive subject matter will be carried out. The area of interest can be an a priori, user-defined area or an ad-hoc area generated by the system.

(14) For a priori defined areas, an area of interest can correspond to existing, predefined boundaries that can be physical (e.g., the physical boundaries of a road or a beachfront up to the water, the structural boundaries of a building, etc.), non-physical (e.g., a geographical boundary, geo-political boundary (e.g., a country border, an embassy's territory, etc.), geofence, territorial boundary (e.g. real-estate property boundaries, etc.), jurisdictional boundary (city, state, town, county, etc.), or other boundary defined by limits or borders not constrained to a physical demarcation) or a combination of both (e.g., a section of a room inside a building defined by some of the walls in the room and also a user-defined boundary bisecting the room, a subway station platform area defined by user-set boundaries and the subway tracks, a park's boundaries having state-defined boundaries over land on some of the sides and a natural boundary such as a river on the remaining side, a landmark whose boundaries are defined by the structural borders of the landmark itself on some sides and by surrounding gardens or walkways on remaining sides, etc.). Thus, it is contemplated that areas of interest can be as large as a state, city, county, town, national park, etc., or as small as a section of a room inside a building or house.

(15) In embodiments, a user can set an area of interest by selecting a pre-existing area from a map, blueprint, etc. For example, selecting a landmark as the area of interest would incorporate the boundaries of the landmark as denoted on a map. Likewise, selecting a floor of a building as the area of interest would include the floor as denoted in the official floor plan or blueprints for the building. The user can also set an area of interest by manually setting and/or adjusting the desired boundaries of the area of interest on a graphical user interface. In one example, the user can select a point or coordinate on a rendered digital map and extend the area of interest radially outward from the point. In another example, the user could denote the area of interest on a map, blueprint, floor plan, etc., by manually drawing the line segments corresponding to the boundary or as a bounding box. A user can access map generation engine **102** via a user interface that allows the user to manually generate, via the graphical user interface, and/or adjust the area of interest. Suitable user interfaces include computing devices (e.g., smartphones, tablets, desktop computers, servers, laptop computers, gaming consoles, thin clients, fat clients, etc.) communicatively coupled to the map generation engine **102** and other system components. These user interfaces can include user input devices such as a keyboard, mouse, stylus, touchscreen, microphone, etc. to input data into the user interface and output devices such as screens, audio output, sensory feedback devices, etc. to present output data to the user.

(16) Contemplated areas of interest include all suitable interior and outdoor settings. Examples of indoor settings can include a casino, an office space, a retail space, an arena, a school, an indoor shopping center, a department store, a healthcare facility, a library, a home, a castle, a building, a temporary shelter, a tent, an airport terminal, a submarine, or any other interior setting. Examples of outdoor settings can include a stadium, a park, a wilderness area, an arena, a road, a field, a route, a highway, a garden, a zoo, an amusement park, the outside of an airport, the outside of a cruise-ship, a sightseeing tour, a rooftop or any other outdoor setting.

(17) In embodiments, the map generation engine **102** of system **100** can generate an ad-hoc area of interest based on a number of devices detected in a particular area at a particular time. To do so, the map generation engine **102** can receive position data corresponding to a plurality of user devices, via clustering or other statistical algorithms, determine that a threshold number of devices are within a certain distance of one another and/or within or passing through a monitored space or point within a designated area (e.g., a train platform, a point in an airport terminal hallway, etc.). If the threshold is met, the map generation engine **102** can then generate the area of interest such that the area encompasses the cluster and, optionally, an additional distance from the cluster. In a variation of these embodiments, the ad-hoc area of interest can be an a priori area of interest modified according to the number of devices present as well as other factors such as modifications to the real-world area or structure, modifications to traffic patterns, etc. For example, for a train platform corresponding to an a priori defined area of interest, the map generation engine **102** can be programmed to modify the boundaries of the defined area of interest based on the length of the next train (since people are not going to gather around to enter cars beyond the last car in a particular train). One should appreciate that although the area of interest corresponds to a physical location, within the disclosed system the area of interest comprises a data structure that includes attributes and values that digitally describe the area of interest. Thus, the area of interest can be considered a digital model or

object of the area of interest in a form processable by the disclosed computing devices.

(18) One should appreciate that where area data obtained of different modalities are available, especially where there is a vast amount of area data available, a system of the inventive subject matter could operate with an increased level of accuracy (e.g., accuracy with respect to map dimensions, point of view origins, field of views, views, objects within a view of interest, locations, measurements of six degrees of freedom, etc.). Thus, and viewed from another perspective, the AR management engine **130** could be configured to obtain or otherwise utilize area data comprising different modalities and different views of every portion of the area of interest. This data could be used to obtain an initial map having increased accuracy, and to generate a tile map having increased accuracy such that a user's device could be configured to obtain AR content objects **134** and instantiate those objects at the precise moment (e.g., precise location, precise positioning of the device, etc.) they are intended to be presented.

(19) AR content objects **134** can be data objects including content that is to be presented via a suitable computing device (e.g., smartphone, AR goggles, tablet, etc.) to generate an augmented-reality or mixed-reality environment. This can involve overlaying the content on real-world imagery (preferably in real-time) via the computing device, such that the user of the computing device sees a combination of the real-world imagery with the AR content seamlessly. Contemplated AR content objects can include a virtual object, chroma key content, digital image, digital video, audio data, application, script, promotion, advertisements, games, workflows, kinesthetic, tactile, lesson plan, etc. AR content objects can include graphic sprites and animations, can range from an HTML window and anything contained therein to 3D sprites rendered either in scripted animation or for an interactive game experience. Rendered sprites can be made to appear to interact with the physical elements of the space whose geometry has been reconstructed either in advance, or in real-time in the background of the AR experience.

(20) In some embodiments, AR content objects **134** could be instantiated based on object recognition and motion estimation within an area of interest or movement to or from areas of interest. In such embodiments, it is contemplated that the device configured to obtain AR content objects **134** could comprise at least a camera and a gyroscope. Suitable techniques for image recognition can be found in, among other things, co-owned U.S. Pat. Nos. 7,016,532, 8,224,077, 8,224,078, and 8,218,873, each of which are incorporated by reference herein. Suitable techniques for motion estimation can be found in, among other things, “3-D Motion Estimation and Online Temporal Calibration For Camera-IMU Systems” by Li, Mingyang, et al; “Method For Motion Estimation With A Rolling-Shutter Camera” by Mourikis, Anastasios, et al; and “Method For Processing Feature Measurements In Vision-Aided Inertial Navigation” by Mourikis, Anastasios; each published by the Department of Electrical Engineering, University of California, Riverside and all of which are incorporated by reference in their entirety.

(21) One should also appreciate that there could be a hierarchy of modalities with respect to precision and error tracking, and that this hierarchy could be determined by one or more users, or by the system. Thus, a system manager recognizing that one modality is more reliable than others could cause the map generation engine to prioritize data according to their modality where there is a conflict. For example, audio data in area database **110** could describe a layout of a record store (e.g., distances, signs, merchandise, etc.), while video data in area database **110** could include footage of the record store that conflicts with the audio data. It is contemplated that the audio data could be prioritized over the video data (e.g., based on a time the data was captured, etc.), or that the video data could be prioritized over the audio data (e.g., based on a general increased level of reliability against human error, etc.). The initial map or other map could be generated based on both of the audio and video data, except that the audio data, for example, could be ignored to the extent that it conflicts with the video data.

(22) One possible technology that could be utilized by a system of the inventive subject matter is fingerprint-based techniques using an existing infrastructure. For example, as a user navigates an area of interest with a device having one or more sensors, the device could identify access points throughout various portions of the area, and determine available networks (e.g., wireless networks, etc.), a received or detected signal strength (e.g., WiFi signal, cellular network signal, etc.) at a particular time, or to obtain information related to what a user could observe, hear or otherwise experience in the portions at various times.

(23) It is also contemplated that a series of initial maps could be generated for an area of interest, wherein the initial maps use different portions of the available area data. In such embodiments, the initial map that is obtained by a given AR management engine **130** could be determined based on the sensor(s) being used by the device configured to obtain and instantiate AR content objects. For example, where a user of a record store specific application is navigating the record store using a mobile phone capturing voice inputs of the user, it is contemplated that the initial map obtained by the AR management engine is one generated using more audio data relative to other initial maps of the area. As another example, where the user is navigating the record store using a mobile phone capturing video or image inputs captured by the user, it is contemplated that the initial map obtained by the AR management engine is one generated using less audio data relative to other initial maps of the

area.

(24) Initial map **118** can comprise a CAD drawing, a digital blueprint, a three-dimensional digital model, a two-dimensional digital model or any other suitable digital representation of a layout of an area of interest. In some embodiments, the initial map **118** could comprise a digital or virtual construct in memory that is generated by the map generation engine **102** of system **100**, by combining some or all of the image data, video data, signal data, orientation data, existing map data (e.g., a directory map of a shopping center already operating, etc.) and other data.

(25) FIG. 2 is a schematic showing a generation of an initial map **118** of an area of interest. As mentioned above, an initial map **118** of an area can comprise a digital or virtual construct, and can be generated via map generation engine **102**. In the example shown, map generation engine **202** (corresponding to map generation engine **102** of FIG. 1) is coupled with user interface **200A**, sensor **200B** (i.e., device comprising a sensor), and sensor **200C** via network **205**, and configured to generate an initial map based on data (e.g., image data **200A**, video data **202B**, signal data **202C**, etc.) acquired from one or more of user interface **200A**, and sensors **200B** and **200C**.

(26) User interface **200A** could be used by one or more users to transmit data related to an area of interest to map generation engine **202**. The following use case provides an example of how various users could cause area data to be transmitted to map generation engine **202**. Abigail, Bryan and Catherine have each posted various images and videos of the Los Angeles Airport (LAX) on various social networking websites (e.g., Facebook®, MySpace®, Twitter®, Bebo®, Tagged®, Flixster®, Netlog®, etc.). Abigail, visiting from Australia, posted several videos on her Twitter® page arriving and departing from the Tom Bradley International Terminal of LAX. Bryan, visiting New Mexico, posted several images on Facebook® taken from Terminal 1 of LAX. Catherine, picking Bryan up from the airport, posted a video captured while she circled LAX waiting for Bryan to arrive, as well as several photographs taken with Bryan in the parking structure of LAX. David, a system manager responsible for creating a mobile app targeting LAX visitors, obtains the images and videos from Abigail, Bryan and Catherine's profiles, and transmits them to map generation engine **202** via user interface **200A**. It should also be appreciated that map generation engine **202** could be coupled with various social networking websites or other sources and automatically obtain area data from those sources, for example, using an Internet bot.

(27) David has also set up various devices throughout LAX having sensors (e.g., **200B** and **200C**) that captured image data and video data throughout LAX, as well as activity information to determine, or allow a determination of areas having high, medium or low traffic. Area data is transmitted from these devices to map generation engine **202** via network **205**. Once the map generation engine **202** receives adequate data, it is contemplated that an initial map **218** of LAX could be generated. The initial map **218** could be generated manually (e.g., a user could utilize the area data to create the initial map, etc.), or by the map generation module **202** itself. For example, the map generation engine **202** can comprise a data compiling module that sorts the area data into groupings (e.g., based on location, based on popularity, etc.), and a mapping module that uses the sorted area data to automatically generate an initial map **218**. In some embodiments, it is contemplated that the initial map could provide information not only related to a layout, but also to traffic, popularity, time, or other characteristic related to behavior. Once the initial map is finalized, the map generation engine **202** could transmit the initial map **218** to area database **210** for storage via network **215**.

(28) In another example, the sensors (e.g., sensors **200B**, **200C**) can be placed on a drone or remote-controlled robot that can be programmed to travel within the area of interest to gather the data, and transmit it to map generation engine **202**.

(29) To generate the initial map **218**, the mapping module of map generation engine **202** can employ a "structure from motion" module capable of generating a 3D map of the geometry depicted in images and thus construct a 3D model of the area of interest. To create a 2D blueprint or floor plan, the map generation engine **202** can "flatten" the constructed 3D model.

(30) During the flattening, the map generation engine **202** can label certain geometric features of interest within the 3D model (e.g., doors, windows, multi-level spaces or structures, overpasses and/or underpasses in a building, etc.) via classifiers trained offline in advance of the flattening process. These classifiers can be mapped to corresponding geometric features of interest via a recognition of these features in the 3D model and/or the image data used to generate the 3D model using image recognition techniques.

(31) Examples of suitable "structure from motion" and other techniques usable in generating the initial map (and/or gathering the data to be used in the generation of the initial map) can include those discussed in U.S. pre-grant publication number 2013/0265387 to Jin, entitled "Opt-Keyframe Reconstruction From Robust Video-Based Structure From Motion" and published Oct. 10, 2013; U.S. pre-grant publication number 2014/0184749 to Hilliges, et al, entitled "Using Photometric Stereo For 3D Environment Modeling" and published Jul. 3, 2014; U.S. pre-grant publication number 2012/0229607 to Baker, et al, entitled "Systems and Methods for Persistent Surveillance And Large Volume Data Streaming" and published Sep. 13, 2012; all of which are incorporated

herein by reference in their entirety.

(32) In embodiments, the initial map **218** can be generated using depth sensing, perhaps through LiDAR techniques combined with image recognition techniques. Suitable LiDAR techniques include those employed by the Zebedee indoor mapper developed by CSIRO and GeoSLAM. Depth sensing can also be achieved through image-based analysis such as those disclosed in U.S. pre-grant publication number 2012/0163672 to McKinnon, entitled “Depth Estimate Determination, System and Methods” and published Jun. 28, 2012, which is incorporated by reference in its entirety, as well as the references discussed above.

(33) These techniques allow for the generation of initial map **118** based on data gathered from a single pass through of the area of interest, such as the aforementioned drone or remote-controlled robot or drone.

(34) In embodiments, it is contemplated that an initial map **118** can be generated or modified manually via a user interface. For example, one or more users can view a plurality of images showing different portions of an area of interest and manually create a CAD drawing based upon the various images. As another example, it is contemplated that one or more users could utilize software that associates different images and generates area maps using portions of some or all of the images and possibly other sensor data (e.g., audio, notes, etc.).

(35) Based on an applicable initial map **118A** (applicable to a selected area of interest) and optional ancillary area data (e.g., image, video, audio, sensor, signal or other data, etc.), the AR management engine **130** can derive a set of views of interest **132** related to the area of interest.

(36) A view of interest **132** is a digital representation of a physical location in real-world space that is to be enabled with AR content. Thus, the view of interest **132** can be considered to be a view or perspective of a view representative of where users would, should, or could be looking while navigating through various portions of the area of interest, for the presentation of AR content.

(37) A view of interest **132** can comprise one or more individual views within an area of interest, from a set of defined perspectives (e.g., from areas near defined points of origin and/or area with a tile, as discussed in further detail below) within the area of interest. The view of interest **132** can include a set of contiguous views within the area of interest, or a set of discontinuous views. Thus, for example, a view of interest **132** in an area of interest can include a view of a section of the area of interest that is in front of a user (and thus, visible to the user at that particular point in time), and another view that is behind the user, across from the first view (and thus, only visible to the user when the user turns around).

(38) The view of interest **132** can be a data construct that typically includes, among other things, one or more point of view origins, at least one field(s) of interest leading to a view, objects of interest within a view, and descriptors associated with objects of interest. The view of interest **132** can also include data associated with one or more of an owner, metadata, a direction (e.g., a vector, an angle, etc.), an orientation (e.g., pitch, yaw, roll, etc.), a cost, a search attribute, or any combination or multiples thereof.

(39) In embodiments, views of interest **132** within an area of interest can be selected and derived entirely by AR management engine **130**. FIG. 3 is an illustrative example of an exemplary process of these embodiments.

(40) At step **310**, the AR management engine **130** obtains the initial map **118A** and area data associated with the area of interest. As described above, this area data can be image data, video data, audio data, sensor data, signal data, and any other data associated with the area of interest.

(41) At step **320**, the AR management engine **130** can employ one or more data analysis and recognition techniques on the area data to assess the characteristics of the area of interest environment and recognize objects in the area of interest environment, as appropriate for the modalities of the area data.

(42) For example, for image or video data, the AR management engine **130** can employ image recognition techniques, such as those mentioned herein, to recognize and identify real-world objects within the area of interest.

(43) For audio data (either audio-only, or accompanying video data), the AR management engine **130** can employ audio recognition and analysis techniques to identify the acoustic characteristics of the environment, locations of sources of sound (e.g., locations of speakers or other audio output devices, sources of environmental noise, etc.), and/or identification of audio (e.g., for music, identify songs, genres, etc.; for sounds, identify the type of sounds, the source producing the sound, etc.).

(44) Sensor data can include temperature sensor data, air pressure sensor data, light sensor data, location-sensor data (e.g., GPS or other location- or position-determination system data), anemometer data, olfactometer data, etc. Correspondingly, the AR management engine **130** can determine the temperature, air flow characteristics, lighting characteristics, smell characteristics and other environmental characteristics for various locations within the area of interest.

(45) Signal data can correspond to data within and also about signals from routers, signals from cellular transmitters, signals from computing devices (e.g., desktop computers, laptop computers, smartphones, tablets, gaming consoles, remote controls, etc.), broadcast signals (e.g., over-the-air television or radio broadcasts), near-

field communication devices, or other emitters of wireless data carrier signals. Types of signals can include WiFi signals, cellular signals, mobile hotspot signals, infrared signals, Bluetooth® signals, NFC signals, ultrasound signals, RFID signals, or any other detectable data carrier signal. The signal data itself can include information such as identification of emitting device, identification of standard(s)/protocol(s), network location information (IP address, etc.), physical location information of emitter, etc. The AR management engine **130** can analyze the signal data (corresponding to the signals themselves and/or the information carried by the signal) to determine the location(s) of various signal emitters with the area of interest, the signal strength of the various signals within the various parts of the area of interest, potential sources of interference, relatively strong/weak areas of various signals, data transmission speeds, etc.

(46) The recognized objects and characteristics of the environment can be associated with particular locations within the area of interest by correlating the area data with the initial map **118A** based on one or more of the location information (e.g., GPS or other location-sensor information) and location information associated with image data (e.g., depth map information or other information indicative of depth in image).

(47) At step **330**, the AR management engine **130** can obtain descriptors for the recognized objects within the area of interest. The descriptors can be SIFT descriptors, FAST descriptors, BRISK descriptors, FREAK descriptors, SURF descriptors, GLOH descriptors, HOG descriptors, LESH descriptors, etc. In embodiments, the AR management engine **130** can obtain the descriptors from a descriptor database corresponding to various objects capable of being recognized. In embodiments, the AR management engine **130** can derive the descriptors itself, according to known techniques.

(48) At step **340**, the AR management engine **130** can associate at least some of the recognized objects within the area of interest with AR content types or categories. These recognized objects can be considered to be potential “attachment points” for AR content. These attachment points can be identified as potential objects to which AR content objects can be associated within the area of interest to varying levels of specificity or granularity. In other words, the “type” of AR content object identified as applicable to the attachment point can be of a variety of levels of generality or granularity. Certain attachment points can be theme- or topic-independent, merely identified as suitable object to which content can be attached or associated. Examples of these types of attachment points can be recognized billboards, large sections of wall, plants, floor patterns, signage, logos, structural supports, etc. Other attachment points can be topic- or theme-specific to various levels of specificity. For example, if a car is recognized within the area of interest, the AR management engine **130** is programmed to associate the recognized “car” to AR content object categories associated with cars. However, the “car” category can have further subcategories of “sports car”, “SUV”, “luxury car”, etc. Thus, the “car” can be associated with AR content object(s) from one or more applicable sub-categories. In embodiments, the association of step **340** can be based on the descriptors obtained in step **330**. In embodiments, the descriptors of step **330** can correspond to categories of recognized objects on their own, and thus steps **330** and **340** are effectively merged into a single step.

(49) In embodiments, the associations made by the AR management engine **130** can be based on the categorization of the recognized object according to the recognition technique employed.

(50) In embodiments, the associations can be a pre-set association set by system administrators. Thus, the associations can be such that when a “car” is recognized, the AR management engine **130** associates the “car” with AR content objects of the “car” type. This can include associating the recognized “car” only with “car”-type AR content objects, thus ignoring other potential AR content objects that would otherwise be similarly associated with the car.

(51) At step **350**, the AR management engine **130** generates the one or more views of interest **132** for the area of interest based on the initial map **118A** and the area data. To determine what part of the area of interest (reflected in the initial map **118A**) will constitute a view of interest **132**, the AR management engine **130** analyzes the distribution (e.g., density, layout, etc.) of recognized or recognizable objects within the initial map **118A**, including the recognized objects from the perspective of possible point of view origins. The analysis can correspond to a cluster analysis of recognized objects within a particular spatial relationship of one another, and also to possible point-of-view origins. The point-of-view origins correspond to various points within the area of interest from which a user will view a view of interest **132** or part of a view of interest **132**. Thus, the location, size and shape of a view of interest can be determined based on having a certain amount (minimum or maximum) of recognized objects within the view of interest, a certain density of recognized objects, a certain layout, etc. For example, the system could assign a point in space for each recognizable object. The point in space might be the centroid of all the image descriptors associated with the recognized object as represented in 3-space. The system can then use clusters of centroids to measure density. In embodiments, the point-of-view origin can correspond to the point of origin of the area data such as image keyframe data was captured during the initial map-making process.

(52) In embodiments, the views of interest **132** can be based on area data of one or more modalities of the area data as applied to the initial map **118A**. For example, candidate views of interest **132** for an area of interest can be limited to those sections of the area that were captured by visual data (i.e., image or video data). In further embodiments, candidate views of interest can be based on one or more of the area data as applied to initial map **118A** and modified by additional area data. For example, a candidate view of interest **132** for an area can be initially defined by image or video data gathered (that directly show potential views of interest **132** captured visually), which can be expanded or constricted or even eliminated as a candidate based on sound, temperature or other sensor data. In this example, sound data could indicate that there is consistent background audio noise in the particular section of the area of interest being considered, thus being a less desirable candidate for certain AR content objects having audio and also indicative of the fact that people passing through might move quickly and be less inclined to stop and consume presented content.

(53) Based on the initial map **118A** as well as the area data, the AR management engine **130** can determine potential fields of interest for each view of interest **132**. A field of interest can be considered to be the perspective or field of view that leads to a view of a part of or all of a view of interest **132**. In other words, the field of interest can be considered to be a potential field of view of a user (i.e., the user's visible area as seen through a display device on a smartphone or other computing device that displays a live video feed, via AR goggles or glasses, etc.) that would cause the user to see a particular view within a larger view of interest **132** at any given time. Thus, if a view of interest **132** includes a section of the area of interest in front of the user as well as behind the user, the view of interest **132** is considered to have at least two fields of interest—one that captures the view of interest **132** portion in front of the user (which would be considered a first view within view of interest **132**), and another that corresponds to the field of view of the portion of the view of interest **132** behind the user, requiring the user to turn around to see it (which would be considered a second view within view of interest **132**). Additionally, the fields of interest can account for obstructions and other obstacles that would interfere with the user's view of some or all of a view of interest **132**.

(54) It should be appreciated that the area data gathered to generate views of interest **132** could be derived in any commercially suitable manner (e.g., crowd sourced using ambient collection, GoPro® or other suitable technologies, using available image, video or other data, customized through paid-for data, automated drones, etc.). The following use case illustrates one method in which view(s) of interest could be derived in a customized manner. A system manager hires various shoppers (Martin, Nick, Mei and Bob) at The Grove® shopping mall to videotape their shopping experience. Each shopper is to wear a video capturing device (e.g., attached to the shopper's hat, shirt, etc.) while they go about their usual shopping experience. In some embodiments the shoppers could be selected based on a similarity in interests or other characteristics (e.g., age, gender, income, demographic, psychographic, employment, sexual orientation, etc.). This could be advantageous where a system wishes to cater to a selected group of people (e.g., high school kids from affluent neighborhoods, etc.) In this example, Martin, Nick, Mei and Bob are selected because of the dissimilarity in interests, age and gender. This could be advantageous where a system wishes to cater to a wide range of people regardless of their interests.

(55) Martin and Nick each wear their video cameras on their hat as they navigate The Grove® together. Because their interests are widely varied, the field of views and objects that are captured from the same or substantially similar point of view origin could be very different. For example, while Martin and Nick could each be standing two feet apart from each other next to the fountain at the Grove®, Martin could be capturing video data including the sky and the movie theatre, while Nick could be capturing video data including the Nordstrom®, the Farm® restaurant, and Crate and Barrel®. Meanwhile, Bob could be sitting at the Coffee Bean® capturing video data including various portions of the farmer's market neighboring the Grove®, while Mei could be inside The Children's Place® shopping for her kids and capturing video data including various portions of the store.

(56) Based on the initial map **118A** and the video data captured by Martin, Nick, Bob and Mei, the AR management engine **130** could derive a set of views of interest **132**. It is contemplated that some or all of the view of interest information could be derived by the AR management engine **130**. Alternatively or additionally, some or all of the view of interest information could be derived elsewhere and obtained by the AR management engine **130** (e.g., descriptor information, etc.).

(57) While the above example focuses on obtaining views of interest **132** from specifically selected individuals, it should be appreciated that views of interest **132** could be obtained using any suitable method. For example, images could be taken from one or more specifically adapted vehicles, robots or other devices and stitched together to produce a segmented panorama or high resolution image. Each device could be configured to obtain image data from various angles at different heights. Additionally or alternatively, the devices could include 3G, 4G, GSM, WiFi or other antennas for scanning 3G, 4G, GSM, WiFi or other signals and hotspots. As another example, the system could leverage asset tracking (e.g., RFIDs, etc.) or crowd sourcing technologies to obtain area data from users who do not have a specific goal of providing area data for purposes of generating initial and

tessellated maps.

(58) In embodiments, views of interest **132** can be selected by human users (e.g., a system administrator, advertiser, merchant, etc.) for derivation by the AR management engine **130**. In these embodiments, the area data (such as the image data corresponding to the views) can be presented to users from which the human user(s) can select a corresponding view of interest **132**. For example, an advertiser can be shown image data of various sections of the area of interest. From these images, the advertiser can select one or more images showing a particular section of the area of interest that the advertiser wishes to use to present advertisements to users. The AR management engine **130** can then generate the view of interest **132** corresponding to the selected section based on the initial map **118A** and the area data associated with the selected section of the area of interest.

(59) System **100** can also comprise an object generation engine **104**, which could obtain a plurality of content objects (e.g., image content objects **122**, video content objects **124**, audio content objects **126**, etc.) from one or more users or devices, and transmit the objects to AR content database **120** via network **115**. For example, a system manager could upload AR content obtained from various advertisers who wish to advertise a good or service to people visiting or residing in an area of interest (such as a shopping mall). The system manager could also include ancillary information such as advertiser preferences, costs, fees, priority or any other suitable information. The AR content objects and the ancillary information could be stored in the database, and could be associated with various descriptors (e.g., SIFT, FAST, BRISK, FREAK, SURF, GLOH, HOG, LESH, TILT, etc.) stored in database **105** by one or both of the object generation engine **104** or the AR management engine **130**.

(60) Once the views of interest **132** have been derived, and AR content objects have been generated, AR management engine **130** could obtain a set of AR content objects **134** (e.g., from the AR content database **120** via network **135**) related to the derived set of views of interest **132**. It should be appreciated that the set of AR content objects **134** could be obtained in any suitable manner, including for example, based on a search query of AR content database **120** (e.g., a search for AR content objects **134** in database **120** that are associated with one or more descriptors that are associated with one or more views of interest **132**, etc.), based on a characteristic of the initial map **118A** (e.g., dimensions, layout, an indication of the type of area, etc.), based on a user selection, recommendation or request (e.g., by an advertiser, merchant, etc.), or based on a context of an intended use of a user (e.g., based on what activities a user wishes to capture (e.g., shopping, educational, sightseeing, directing, traveling, gaming, etc.).

(61) As a function of at least one of the AR content objects **134** and the set of views of interest **132**, AR management engine **130** establishes AR experience clusters **136** within initial map **118A** or as a new map.

(62) For example, AR experience clusters **136** can be established to include one or more point of view origins from which objects of interest could be viewed based on a density of AR content objects **134** associated with the point of view origins of the various views of interest **132**. Viewed from another perspective, each experience cluster **136** can include point of view origins such that the point of view origin(s) in each cluster correspond to a substantially equal percentage (e.g., deviations of $\leq 5\%$, $\leq 3\%$, $\leq 1\%$, etc. from each of the other clusters) of the total AR content objects **134**. As another example, each experience cluster could include point of view origins such that the point of view origin(s) in each cluster correspond to a substantially equal percentage (e.g., deviations of $\leq 5\%$, $\leq 3\%$, $\leq 1\%$, etc. from each of the other clusters) of at least one of the following: video content objects **124**, image content objects **122**, and audio content objects **126**. As yet another example, one or more of the experience clusters could include point of view origin(s) that are associated with only a few AR content objects (e.g., less than 10, less than 5, less than 3, 1, etc.), for example where an advertiser has paid a premium to obtain the exclusivity, whereas the remaining experience clusters could include point of view origins that are associated with more AR content objects **134** (e.g., at least 50% more, at least 100% more, at least 200% more, at least 300% more, at least 400% more, at least 500% more, etc.). One should appreciate that a cluster could be established based on any suitable parameter(s), which could be established manually by one or more users, or automatically by a system of the inventive subject matter.

(63) It should be appreciated that, for point of view origins of various distances, a same section of the area of interest can have multiple views of interest **132** and/or multiple experience clusters **136**. For example, an area of interest has a wall with a number of advertisement poster objects that have been recognized and potentially can be linked to AR content objects. As a user gets closer to the wall, there will be less posters appearing in the user's field of view. Conversely, as the user gets farther away from the wall, there will be more posters appearing within the user's field of view. In this example, multiple view of interest **132** can be derived to account for the differences in the amount of potential recognized attachment points (the recognized poster objects) at different view point-of-origin distances.

(64) Based on the established AR experience clusters **136**, the AR management engine **130** could generate an area tile map **138** of the area of interest. The tile map **138** could comprise a plurality of tessellated tiles covering the area of interest or portion(s) thereof. Depending on the parameters used to establish the AR experience clusters

136, the area tile map **138** could comprise a regular tessellation, a semi-regular tessellation, an aperiodic tessellation, a Voronoi tessellation, a Penrose tessellation, or any other suitable tessellation. The concepts of establishing experience clusters and generating tile maps are discussed in further detail below with FIGS. 5 and 6. (65) The AR management engine **130** in some embodiments could be coupled with a device **140** (e.g., cell phone, tablet, kiosk, laptop computer, watch, vehicle, etc.) via network **145**, and configure the device to obtain at least a portion of the subset of the AR content objects depending on at least one of the following: the location of the device within an area of interest (e.g., within a location represented within a tile of area tile map **138**, etc.), and the objects viewable by the device or a user of the device. For example, it is contemplated that an area tile map **138** could comprise a first tile that is representative of portions of the area map corresponding to point of view origins located next to the fountain and next to Crate and Barrel® at the Grove®. The area tile map **138** could also comprise a second tile bordering a portion of the first tile, which is representative of portions of the area map corresponding to point of view origins located next to the Coffee Bean® and next to the Children's Place® at the Grove®. As a user carrying device **140** comes closer to the portion of the Grove® represented by first tile of the map from a portion represented by the second tile of the map, the user device **140** can be auto-populated with a subset of AR content objects **134A** associated with the first tile map. When the user walks to or near a point of view origin and captures image data related to an object of interest within a view of interest, system **100** could associate the object of interest with one or more of the subset **134A** (e.g., based on a descriptor or other identification, etc.) and instantiate them for presentation to the user.

(66) FIG. 4 is a schematic showing the generation of an AR content database **420**. Object generation engine **404** is coupled with descriptor database **405**, AR content database **420**, and user interfaces **400A**, **400B**, and **400C** via networks **425**, **415** and **405**, respectively. The following use case provides an example of how AR content objects can be associated with one or more descriptors related to objects viewable from an area of interest, and stored in an AR content database **420** for use by an AR management engine of the inventive subject matter.

(67) In the example shown in FIG. 4, descriptor database **405** comprises information related to various objects viewable from within the Aria® Hotel and Casino in Las Vegas (the area of interest). Descriptor database **405** could comprise, among other things, object image data, descriptors associated with the image data, and information relating to the device capturing the object image data (e.g., translation data (heaving, swaying, surging), rotation data (e.g., pitching, yawing, rolling), distance, clarity, brightness, location, etc.). While this example is generally related to associating a descriptor with an object generally (without consideration for a specific portion of the object or the device capturing data related to the object, etc.), it should be appreciated that obtaining information related to the data capturing device at the time of capture could be valuable in providing an increased level of accuracy as an object captured from one position, view or angle could be associated with AR content objects that are different from the same object when captured from a different position, view or angle.

(68) For example, descriptor database **405** could comprise descriptor set A (**405A**) including SIFT descriptors associated with an image of the host desk of the Aria® poker room, descriptor set B (**405B**) including SIFT descriptors associated with an image of the host desk of the Aria® buffet, and descriptor C (**405C**) including SIFT descriptors associated with an image of the Aria® concierge desk.

(69) Alex, the general manager of Aria® could use user interface **400A** to transmit content object **422C**, content object **424A** and content object **426A** to object generation engine **404**. Object **422A** comprises an image of Phil Ivey playing Texas Hold'em in the Aria® poker room to generate interest in the poker room, object **424A** comprises a video of a model getting a massage at the hotel to advertise the hotel amenities, and object **426A** comprises an audio of the lunch menu to assist the visually impaired. Brandi, an advertising executive could use user interface **400B** to transmit content object **422B** to object generation engine **404**. Content object **422B** comprises an image of an advertisement for skydiving classes located right off the Las Vegas strip. Carina, a system manager responsible for creating a mobile app for Aria® visitors, could transmit content object **422C**, an image of a map of the Aria® hotel, to object generation engine **404**, and could also associate the various descriptors **405A**, B and C with one or more content objects. In the example provided, Carina associates content objects **422A** and **424A** with descriptor **405A**, content objects **422B** and **426A** with descriptor **405B**, and content objects **422C** and **424B** with descriptor **405C**. This association could be based on any suitable parameters as determined by one or more users or the object generation engine itself.

(70) Object generation engine **404** could transmit the image AR content objects **422**, video AR content objects **424**, audio AR content objects **426**, and optionally the associated descriptors to AR content database **420** via network **415**.

(71) In embodiments, suitable content objects AR content objects **422** can additionally be identified via the content types associated with the recognized objects at step **340**. Thus, for a particular recognized object, the AR content objects **422** can be selected based on the descriptor of the object itself, as well as according to the categorization or other classification associated with the object.

(72) FIG. 5 is a schematic of an AR management engine 530 generating area tile maps 538 and 538T. AR Management engine 530 comprises, among other things, an initial map 518A, view(s) of interest 532 (e.g., point of view origin data, field of interest data, etc.), descriptors associated with view(s) of interest 533, and AR content object(s) 534. Based on the aforementioned area data and optional other data (e.g., signal data, etc.), AR management engine 430 establishes AR experience clusters A, B, C and D within the initial map 518A as a function of the set of AR content objects 534 and the views of interest 532. In the example shown in FIG. 5, Cluster A comprises a first point of view origin (e.g., a coordinate) having a first field of interest leading to view A. View A could comprise various objects of interest to which descriptors could be associated. Continuing on the example from FIG. 4, the point of view origin associated with view A could comprise the Aria® poker room along with the Aria® poker room host desk to which descriptor 405A is associated. Thus, when a user uses a mobile device to scan view A (or portion thereof) from the related point of view, the user could be presented with content objects 422A and 424A described above.

(73) Furthermore, Cluster B comprises the point of view origins having fields of interest leading to views B and Z; Cluster C comprises the point of view origin having the field of interest leading to view W; and Cluster D comprises the point of view origins having fields of interest leading to views X and Y. Each of clusters B, C and D could include point of view origin(s) having corresponding fields of interest and views including objects of interest. The establishing of clusters could be based on any suitable parameter(s), including for example, the number of objects of interests viewable from a point of view origin, field of view or view, a number of AR content objects associated with objects of interests within a view of interest, a file size of AR content objects of interest within a view of interest, a AR content object type (e.g., image, video, audio, etc.), a number of views of interests viewable from point of view origins within an area of interest, or any other suitable parameter(s). Moreover, any suitable algorithm(s) or method(s) of clustering can be utilized to establish experience clusters, including for example, centroid-based clustering (e.g., k-means clustering, etc.), hierarchical clustering, distribution-based clustering, density-based clustering, or any other suitable algorithms or methods.

(74) Based at least in part on the AR experience clusters established above, area tile maps 438 and 538T (perspective view and top view) could be generated. The area tile maps could comprise a plurality of tessellated tiles covering at least some of the area of interest (e.g., a portion of the Aria® Hotel and Casino, etc.), and one or more of the tiles could be bound to a subset of the AR content objects 534. In the example of FIG. 4, the tessellation comprises an aperiodic tessellation, and is based at least in part on the density of AR content objects associated with each point of view origin. Viewed from another perspective, the number of different AR content objects that could be populated onto a user device and instantiated by a user device in each tile is substantially similar (i.e., within 25% (more preferably within 10%, within 8%, within 5%, within 3%, or within 1%) by number of AR content objects). For example, the point of view origin in Tile 1 is associated with 25 AR content objects, the point of view origins in Tile 2 are associated with 23 AR content objects (15 in view Z, 8 in view B), the point of view origin in Tile 3 is associated with 25 AR content objects, and the point of view origins in tile 4 are associated with 26 AR content objects (1 in view X, 25 in view Y).

(75) It should also be appreciated that a tessellated map could have more than two dimensions of relevance (e.g., at least 3 dimensions, at least 5 dimensions, at least 10 dimensions, at least 25 or even more dimensions of relevance, etc.). Viewed from another perspective, the tessellation could be based not only on a spatial dimension, but could additionally or alternatively be based on a signal strength (e.g., RSS, CSS, WiFi signal strength, cellular signal strength, demographic, etc.) or any other suitable dimension(s).

(76) One should appreciate that a cluster, a view of interest or any portion thereof (e.g., point of view origin, a field of interest, a view associated with a point of view origin, etc.) could be owned and managed by one or more entities. For example, Card Player® magazine could purchase or rent the view of interest comprising view A, and determine what AR content objects are associated with objects viewable from the point of view origin in Cluster A. Moreover, because Card Player® magazine would own and manage the point of view origin, the magazine could modify the field of interest and location or scope of view A if desired. For example, the field of interest could be dynamic in nature, and could include the Aria® poker room host desk during busy hours (e.g., where the room has reached 50%, 70% or even 90% or more of the allowed occupancy), but include a TV screen in the poker room during slow hours in place of the host desk. Thus, a user scanning the host desk during busy hours could be presented with AR content, while a user scanning the host desk during slow hours could be presented with no AR content (or different AR content). Similarly, a user scanning the TV screen during busy hours could be presented with no AR content, while a user scanning the TV screen during slow hours could be presented with AR content.

(77) FIG. 6 is a schematic showing an area tile map 600 having been generated based on experience clusters. In the example shown, a portion of an area of interest is divided into tiles A and B, which were generated based on experience clusters established based on the number of AR content objects associated with each view of interest

in the portion. View of interest W includes a point of view or view-point origin W having two fields of interest (fields of interest W.sub.1 and W.sub.2) facing two different views (views W.sub.1 and W.sub.2). Views W.sub.1 and W.sub.2 each include objects of interest tied to AR content objects (2 content objects and 5 content objects, respectively). View of interest Y includes a point of view origin X having a field of interest X.sub.1 facing view X.sub.1. View X.sub.1 includes objects of interests tied to three AR content objects. View of interest Y includes a point of view or view-point origin Y having two fields of interest (fields of interest Y.sub.1 and Y.sub.2) facing two different views (views Y.sub.1 and Y.sub.2). Views Y.sub.1 and Y.sub.2 each include objects of interest tied to AR content objects (1 content objects and 2 content objects, respectively).

(78) Based on the number of AR content objects tied to each point of view origin of views of interest, experience clusters are generated. Here, the first experience cluster includes point of view origin W, while the second includes point of view origins X and Y, such that the experience clusters of an area of interest (or portion thereof) include substantially the same density of AR content objects by number. Based on these clusters, Tile A is generated including point of view origin W, and Tile B is generated bordering at least a portion of Tile A and including point of view origins X and Y.

(79) When a user navigating the real world area of interest gets close enough to a portion represented by Tile A (e.g., within 50 feet, within 25 feet, within 10 feet, within two feet, within one foot, etc. of any portion of tile A), it is contemplated that the user's device could be auto-populated with the 7 AR content objects bound to view of interest W. When the user scans view W.sub.1 with a device having a sensor (e.g., camera, etc.), it is contemplated that a system of the inventive subject matter could utilize object recognition techniques to recognize objects of interest within view W.sub.1 and instantiate one or more of the AR content objects associated with the objects of interest. Similarly, when the user scans view W.sub.2, the system could recognize objects of interest within view W.sub.2 and instantiate one or more of the AR content objects associated therewith. When the user navigates closer to Tile B, it is contemplated that the user device will be auto-populated with the AR content objects associated with that tile (e.g., associated with Views W.sub.1, W.sub.1, and W.sub.1, etc.). Additionally or alternatively, it is contemplated that as the user navigates close to Tile B (or any other tile other than Tile A), or as the user navigates away from Tile A (e.g., within 50 feet, within 25 feet, within 10 feet, within two feet, within one foot, etc. of any portion of tile A), the populated AR content objects associated with Tile A could be deleted from the user device automatically or manually.

(80) Viewed from another perspective, a user device in an area of interest could obtain and store AR content objects associated with one or more tiles corresponding to the area of interest. For example, it is contemplated that any time a user device is within 5 feet of a location corresponding with a tile or an area map, the user device will store AR content objects associated with that tile. Thus, if the user device is at a location within 5 feet of two or more tiles, the user device could store AR content objects associated with two or more tiles simultaneously. Moreover, it is also contemplated that the user device, even when located within 5 feet of two or more tiles, could store AR content objects only associated with one of the tiles (e.g., based on a hierarchy, etc.).

(81) It should be noted that while the tiles shown in FIGS. 5-6 show separate, adjacent tiles, it is also contemplated that tiles can overlap within an area of interest. In these embodiments, the AR content objects to be presented can include AR content objects associated with both tiles, either together or with filtering (according to priority, size, etc.). Alternatively, the AR content objects to be presented to the user can (i.e., which tile's AR content objects to show) be determined based on the user's device, a user identification, information about the user (e.g., interests, context, preferences, etc.). Thus, the inventive subject matter is considered to include management of tile overlaps and associated rules for presenting AR content from overlapping tiles. For example, AR content can be prioritized for presentation or rendering based on the fees paid by content providers; a high fee could increase the prioritization of the AR content for example. Other factors that could influence prioritization or placement of AR content from overlapping tiles include personal preferences, demographics, movement of consumer toward or away from tiles, news events, or other factors.

(82) Tiles can be constructed at varying levels of fidelity and resolution to accommodate the various capabilities of several device classes, and tile size can be tuned based on device memory capabilities, network capacity, etc.

(83) It should be appreciated that a point of view origin could comprise any suitable space shape or size, perhaps even geofenced areas or, for example, 10 square feet of a floor, 5 square feet of a floor, 2 square feet of a floor, 1 square foot of a floor, etc. Similarly, a field of interest and/or view of interest could comprise any suitable shape or size.

(84) One should also appreciate that a view of interest could comprise more than a point of view origin, a field of interest, a view associated with a point of view interest, an object of interest, a descriptor set or combinations or multiples thereof. Among other things, a view of interest could comprise an owner (as discussed above), metadata, a direction, an orientation, a cost, a search attribute, or combinations or multiples thereof.

(85) As used herein, a "search attribute" could comprise an object or description that could be used to select a

field of view (or narrow (possible fields of views) to which a user would like to associate content objects with respect to. For example, where an area of interest comprises Magic Mountain®, one possible view of interest could comprise, among other things: the entry point of the line for Batman® the ride as a point of view origin; a field of interest facing 35 degrees above eye level from four feet above the ground, and leading to a view that is ten feet wide (horizontal distance) and four feet long (vertical distance). The view of interest could also comprise a concession stand, a sign pointing to the Green Lantern® ride, and a bench on a hill, each of which could be viewable from the entry point of the Batman® line. In this example, the view of interest could comprise search terms that would assist a provider of AR content objects (or other users) in differentiating this view of interest from others within Magic Mountain, or even from other areas of interests. Exemplary search terms could include, for example, “Batman,” “Green,” “DC®,” “comic,” “superhero,” “rest,” “food,” “drink,” or any other term that describes a characteristic of the view of interest, the area of interest, or the AR content objects that are suitable (or even preferred) for presentation in the view of interest. It is also contemplated that search attributes could be included in a view of interest, which could describe a characteristic of a user experience with respect to the view of interest. For example, a search attribute could comprise an average length of stay of a user within a specified radius of the point of view origin. With respect to the Batman® line entry point, a search attribute could include 20 minutes as the average length of stay of a user within ten feet of the entry point due to the slow pace at which roller coaster ride lines tend to move.

(86) In some embodiments, it is contemplated that a behavior of a user (or user device) could determine some or all of the content that is provided to the user via the user device. For example, and continuing on the example above, a user having a user device in his pocket at Magic Mountain® may stand at a relatively still position for seconds or even minutes at a time. Where the user device scans audio data of announcements over loud-speakers (e.g., safety warnings, etc.) for a pre-determined period of time (e.g., one minute, etc.), this could trigger the user being presented with audio content via the user device advertising Flash® passes, which allow a user to hold his or her place in line electronically. Other examples of behaviors or events that could trigger a provision of content could include, among other things, an interaction with AR content, a comment, a speed of movement, a type of movement, a gesture, a height, or any other suitable behavior or event.

(87) It should also be appreciated that in some embodiments, a system could be configured to allow a user to interact with AR content presented by commenting, tagging, editing, ranking or otherwise modifying or adding to the AR content. This modification or addition could be viewable to all users of the system, a subset of users (e.g., those subscribing to a specific app, friends of the user providing the modification or addition, etc.), or only to be user providing the modification or addition as a reference point.

(88) A contemplated use of a system of the inventive subject matter is to build a scavenger hunt to guide consumers into portions of an area of interest (e.g., a portion of a mall, etc.). Such a system could provide incentives for users to navigate a specific portion of the area, for example, by providing a prize, a reward, a promotion, a coupon, or other virtual item upon an event. The requisite event could comprise simply being located in the portion at any time, or could be more interactive, for example, being located in the portion for a minimum time or a specific time, capturing an image of an object viewable from the portion, making a gesture with the user device in the portion, or any other suitable event.

(89) Location-based services generally rely on one or more sources of information to determine a location of a device. Typical sources include GPS data, Wi-Fi signal strength, or even image features as used in SLAM technologies. However, such techniques often fail in various scenarios. For example, within buildings, GPS signals could be weak or Wi-Fi signals might not be present. Further, in remote locations or natural settings, signals could also be weak or not present. With respect to SLAM-based location technologies, some locations lack sufficient differentiating image features that allow for tracking location of a device. Consider a scenario where a device (e.g., cell phone) is located within a warehouse that has no distinguishing image-based features. That is, there is little distinction from one location to another. Such settings make is very difficult to anchor augmented reality (AR) content within images of the real-world settings.

(90) Another issue with current location-based AR services, especially those based on SLAM, is that they require a user to hold their imaging device (e.g., cell phone, tablet, etc.) up in front of the user. Such a stance can become uncomfortable after a short time for the user. Further, such a stance places the device between the real-world and the user, which restricts the user's interactions with the real-world. A better approach would allow a user to naturally interact with the real-world while their location tracking device is held in a more natural setting.

(91) To address these issues, some embodiments can include encoding a wide area with location information. The location information can take on many different forms including covering surfaces (e.g., walls, floors, ceilings, etc.) with one or more patterns that can be visually observed via an electronic device (e.g., captured via a digital representation of the pattern in the environment such as via image data or video data). The pattern preferably comprises sufficient structure that, when imaged, a location device can observe one or more trackable features

within the pattern. The features can also be bound to locations or used as AR content anchor points. Based on the location information (e.g., feature position, device orientation, etc.), the device is able to determine its location within the wide area.

(92) The pattern can take on many different forms. In some embodiments, the pattern is truly random. For example, the wide area (e.g., warehouse, floors, etc.) can be randomly coated with paint, perhaps infra-red reflective paint. In such a case, the random pattern can then be scanned into a mapping module (e.g., a set of computer-executable instructions stored on non-transitory storage media that, when executed by one or more processors, carry out its described functions) that identifies features in the random paint pattern via one or more image processing algorithms (e.g., SIFT, FAST, etc.) and binds the features to location information. The resulting map can then be deployed in other devices so that they can determine their locations in the environment based on observed features derived from images of the random pattern. The paint pattern can be deployed via a robot, through a suitably configured paint roller, or other means. Further, such a random pattern could be integrated within wall paper, floor tiles, ceiling tiles, or other surface cover at manufacturing time.

(93) Further the pattern could be a natural, existing pattern or texture on a surface. For example, the pattern could comprise wood grain in floor boards (e.g., oak, bamboo, etc.), or concrete. When the setting has acceptable natural textures, a capturing device can be used to map out the area by scanning the all relevant surfaces to build a panoramic map of the locations. Further, the device can be configured to generate a confidence score indicating the acceptability of the natural texture, or other pattern for that matter, on a location-by-location basis.

(94) In other embodiments, the pattern can comprise a generated, pseudo random pattern that covers the wide area. Consider a scenario where a warehouse wishes to encode their warehouse floor with location information. A mapping module can create a pseudo random pattern that creates a feature pattern for the entire space. Perhaps the pattern can be generated from a mapping function based on an initial known seed, which is then concatenated with location information (e.g., X, Y coordinates). The mapping function generates the necessary pattern that should be placed at corresponding X, Y coordinates in the warehouse. For example, each floor tile could be printed with the pattern for that tiles location. The advantageous of such an approach is that the pattern is procedurally generated, which allows devices to derive its location procedurally, assuming it has the initial seed, rather than storing a large, wide area map database.

(95) As more concrete example, consider a case where SIFT is used for deriving features that are then used to determine location of a device. SIFT can have a 128-byte descriptor that represents a feature in an image. In such a case, the pseudo random pattern can be generated by applying an initial seed that is unique to the location to an MD5 hash algorithm (MD5 can generate 128-bit hash values). Once the function is primed with the seed, the X, Y coordinate of a floor tile can be concatenated with the original seed hash multiple times, four times for X and four times for Y. The result of each hash is a 128-bit number. The four hashes for X can be concatenated to form the first 64 bytes of a descriptor and the four hashes for Y can be concatenated to form the last 64 bytes of the descriptor, where the full 128-bytes represents a descriptor corresponding to the floor tile. The 128-byte number can then be considered a SIFT descriptor. For example, if the seed for a location is S and a coordinate is (X, Y), the 128-byte descriptor could be generated as follows: Descriptor bytes 0-15: $X1=MD5(Seed+X)$ Descriptor bytes 16-31: $X2=MD5(X1)$ Descriptor bytes 32-47: $X3=MD5(X2)$ Descriptor bytes 48-63: $X4=MD5(X3)$ Descriptor bytes 64-79: $Y1=MD5(Seed+Y)$ Descriptor bytes 80-95: $Y2=MD5(Y1)$ Descriptor bytes 96-111: $Y3=MD5(Y2)$ Descriptor bytes 112-127: $Y4=MD5(Y3)$

(96) If the hash function is SHA-512, which generates 512 bits of output, then the descriptor could be: Descriptor bytes 0-63: $SHA512(Seed+X)$ Descriptor bytes 64-127: $SHA512(Seed+Y)$

(97) The mapping module uses the 128-bytes descriptor to generate an image pattern that would generate the same descriptor (or a nearest neighbor descriptor for the space) when processed by SIFT. This approach allows for generation of a large number of address spaces. Location can be determined by initially calibrating the device in the local area and using accelerometry to generate a location window. The device can then use the hash mapping functions to generate the descriptors that should be present or observable within the location window.

(98) In some embodiments, a paired dictionary learning process can be used to produce a method of “inverting” any type of image descriptors (SIFT, SURF, HOG, etc.). This can be achieved by keeping original source image patches for all descriptors used to build a dictionary via clustering approaches (K-means, hierarchical K-means, agglomerative clustering, vector quantization, etc.). Once the dictionary is built, it gives a bidirectional mapping from image patch space to descriptor space. From each cluster of descriptors representing a dictionary element, we can obtain an average image patch that would generate a descriptor belonging to that cluster. The chosen size of the dictionary determines the resolution of the mapping between image patches and descriptors, and therefore the size of the available address space of image patches to use when marking up a location. The “inverted” image patterns obtained by this process can be applied to ensure a unique descriptor configuration at each (X,Y) location in a space.

(99) In many embodiments, the mapping module can use a mapping function that is bidirectional in the sense that coordinates generate a desired pseudo random pattern (e.g., a descriptor, feature, keypoint, etc.), and that a descriptor generates a corresponding coordinate. For example, one possible two-way mapping function might include use of a log value. For example, a log (e.g., $\ln(x)$) of the X coordinate of a map location can be taken to generate a value. The value can be the first part of a descriptor (e.g., the first 64 bytes of an SIFT descriptor). The second part of the descriptor could be $\ln(Y)$ after suitable conversion to a 64 byte value. When the descriptor is detected in the field (or a nearest neighbor), the descriptor can be separated into its X and Y parts. The X and Y coordinates can be found by applying an $\exp()$ function to the parts. In some embodiments, X and Y can have only integer values, perhaps as grid locations. Thus, when the $\exp()$ function is applied to observed descriptor values, the resulting nearest integer is likely the proper coordinate.

(100) Yet another possible type of pattern includes a fractal pattern. The pattern can include fractal features that aid in determining location or anchor points at different scales, depth of field, or distances. For example, the pattern could comprise multiple layers where each layer comprises a different color. A first color can be used to generate a fine grained pattern that corresponds to specific location scale (e.g., millimeter, centimeter, etc.). A second pattern having mid-level grained features that provide location scales in an intermediate range (e.g., about 10 cm, 1 foot, etc.). Yet another pattern in a third color having very coarse grained feature might provide location information at a more coarse grain level (e.g., 1 meter, 10 feet, etc.).

(101) Deploying disclosed patterns on a floor surface also provides additional advantages. Users are able to track their location by pointing the image sensors of their devices toward the floor rather than holding the device up. This approach represents a more natural arm position which reduces user fatigue. Further, the user is able to interact with the real-world setting without having the device interposed between the setting the user, thus giving rise to a more natural interaction.

(102) It should be further appreciated that the pattern can be deployed on other surfaces besides the floor. The patterns can be deployed on the walls so that forward facing cameras are able to determine locations of foreground objects (e.g., people, obstacles, machines, etc.) relative to background (i.e., walls) locations. Further, ceilings can also be encoded with patterns (e.g., ceiling tiles, girders, pipes, etc.). In such a case, imaging devices that have a back facing camera, a camera that faces the user, could image the ceiling while also imaging the floor. Thus, the device would be able to derive location information from the ceiling, or both the floor and ceiling. In view that such cameras could have different image capturing resolutions, it is possible that the floor and ceiling patterns could be asymmetric with respect to the location resolving power.

(103) These techniques can be employed to determine a user's position within an area of interest. More specifically, the user's computing device capture a digital representation of a pattern on a surface (e.g., wall, ceiling, floor) within the area of interest, and can determine what tile (for example, Tiles 1-4 of FIG. 5 or Tiles A and B of FIG. 6) a user is currently in by determining the user's position within the area of interest and correlating it to tile information obtained from the system. Then, as described above, the AR content objects can be managed and presented to the user based on their presence within the tile.

(104) As used in the description herein and throughout the claims that follow, when a system, engine, server, device, module, or other computing element is described as configured to perform or execute functions on data in a memory, the meaning of "configured to" or "programmed to" is defined as one or more processors or cores of the computing element being programmed by a set of software instructions stored in the memory of the computing element to execute the set of functions on target data or data objects stored in the memory.

(105) As used herein, and unless the context dictates otherwise, the term "coupled to" is intended to include both direct coupling (in which two elements that are coupled to each other contact each other) and indirect coupling (in which at least one additional element is located between the two elements). Therefore, the terms "coupled to" and "coupled with" are used synonymously.

(106) In some embodiments, the numbers expressing quantities of ingredients, properties such as concentration, reaction conditions, and so forth, used to describe and claim certain embodiments of the invention are to be understood as being modified in some instances by the term "about." Accordingly, in some embodiments, the numerical parameters set forth in the written description and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by a particular embodiment. In some embodiments, the numerical parameters should be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of some embodiments of the invention are approximations, the numerical values set forth in the specific examples are reported as precisely as practicable. The numerical values presented in some embodiments of the invention may contain certain errors necessarily resulting from the standard deviation found in their respective testing measurements.

(107) Unless the context dictates the contrary, all ranges set forth herein should be interpreted as being inclusive

of their endpoints and open-ended ranges should be interpreted to include only commercially practical values. Similarly, all lists of values should be considered as inclusive of intermediate values unless the context indicates the contrary.

(108) As used in the description herein and throughout the claims that follow, the meaning of “a,” “an,” and “the” includes plural reference unless the context clearly dictates otherwise. Also, as used in the description herein, the meaning of “in” includes “in” and “on” unless the context clearly dictates otherwise.

(109) The recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value with a range is incorporated into the specification as if it were individually recited herein. All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g. “such as”) provided with respect to certain embodiments herein is intended merely to better illuminate the invention and does not pose a limitation on the scope of the invention otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the invention.

(110) Groupings of alternative elements or embodiments of the invention disclosed herein are not to be construed as limitations. Each group member can be referred to and claimed individually or in any combination with other members of the group or other elements found herein. One or more members of a group can be included in, or deleted from, a group for reasons of convenience and/or patentability. When any such inclusion or deletion occurs, the specification is herein deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

(111) It should be apparent to those skilled in the art that many more modifications besides those already described are possible without departing from the inventive concepts herein. The inventive subject matter, therefore, is not to be restricted except in the spirit of the appended claims. Moreover, in interpreting both the specification and the claims, all terms should be interpreted in the broadest possible manner consistent with the context. In particular, the terms “comprises” and “comprising” should be interpreted as referring to elements, components, or steps in a non-exclusive manner, indicating that the referenced elements, components, or steps may be present, or utilized, or combined with other elements, components, or steps that are not expressly referenced. Where the specification claims refers to at least one of something selected from the group consisting of A, B, C . . . and N, the text should be interpreted as requiring only one element from the group, not A plus N, or B plus N, etc.

Claims

1. A device capable of rendering augmented reality (AR), the device comprising: at least one sensor, including a location sensor; a display; a non-transitory computer-readable memory storing software instructions; and at least one processor coupled with the non-transitory computer-readable memory, the at least one sensor, and the display; and, upon execution of the software instructions, is configurable to: obtain sensor data from the at least one sensor wherein the sensor data includes a device location obtained from the location sensor; obtain an area of interest via an area database based on at least the device location within the sensor data; access an area tile map of the area of interest, the area tile map represented by a set of tile subareas that includes one or more tessellated tiles from a tessellated tile map; identify a tile subarea from the set of tile subareas based at least in part on the device location relative to one or more locations of tile subareas from the set of tile subareas, wherein the identified tile subarea covers at least a portion of the area of interest, and wherein one or more tessellated tiles within the identified tile subarea are associated with one or more AR content objects; populate the non-transitory computer-readable memory with at least one of the one or more AR content objects associated with the one or more tessellated tiles within the identified tile subarea to thereby accommodate capabilities of the non-transitory computer-readable memory via more accurately instantiating the at least one of the one or more AR content objects in the non-transitory computer-readable memory when the device is at or near the one or more tessellated tiles within the identified tile subarea; and render the at least one of the one or more AR content objects that is associated with the identified tile subarea on the display based on a view of interest.
2. The device of claim 1, wherein the location sensor comprises at least one of the following location sensors: a global positioning system (GPS) sensor, an image sensor, and a wireless sensor.
3. The device of claim 1, wherein the device location includes at least one of the following: a global positioning system (GPS) location, a wireless signal location, a depth of field location, and an image-based simultaneous location and mapping (SLAM) location.
4. The device of claim 1, wherein the at least one sensor further includes one or more of the following: a camera, an accelerometer, and a gyroscope.

5. The device of claim 1, wherein the sensor data further includes at least one of device position data and device orientation data.

6. The device of claim 5, wherein the at least one of the one or more AR content objects is rendered based on the view of interest, wherein the view of interest is derived from at least one of the device position data and the device orientation data.

7. The device of claim 1, wherein the at least one of the one or more AR content objects is rendered on the display as an overlay of an image related to a real-world scene.

8. The device of claim 1, wherein the area of interest comprises one of the following: a landmark, user-defined boundaries, state-defined boundaries, natural boundaries, a city, a country, a business, a shopping center, a warehouse, a stadium, a wilderness area, a road, a garden, a zoo, an amusement park, a beach and a building.

9. The device of claim 1, wherein the area of interest comprises a restaurant.

10. The device of claim 1, wherein the area of interest comprises a coffee shop.

11. The device of claim 1, wherein the area of interest comprises an outdoor setting.

12. The device of claim 1, wherein the device communicatively couples with the area database over a network.

13. The device of claim 1, wherein the at least one of the one or more AR content objects is obtained from an AR content database.

14. The device of claim 13, wherein the device is communicatively coupled with the AR content database over a network.

15. The device of claim 1, wherein the at least one of the one or more AR content objects is rendered relative to a recognized object.

16. The device of claim 15, wherein the recognized object represents an attachment point for the at least one of the one or more AR content objects.

17. The device of claim 16, wherein the attachment point includes at least one of the following: a billboard, a wall, a floor, a plant, a sign, a logo, a landmark, a building, and a car.

18. The device of claim 1, wherein the at least one of the one or more AR content objects comprises a game object.

19. The device of claim 18, wherein the at least one of the one or more AR content objects comprises an interactive game experience.

20. The device of claim 1, wherein the at least one of the one or more AR content objects comprises at least one of the following: a virtual object, a digital image, a digital video, audio data, an application, a script, a promotion, an advertisement, a graphic, a sprite, and an animation.

21. The device of claim 1, wherein the at least one of the one or more AR content objects comprises an exclusive content related to the identified tile subarea.

22. The device of claim 21, wherein the exclusive content related to the identified tile subarea is rendered based on a paid fee.

23. The device of claim 1, wherein the non-transitory computer-readable memory, the at least one processor, the at least one sensor, and the display comprise at least one of the following: a smart phone, augmented reality goggles, a watch, a tablet, a drone, a vehicle, a wearable augmented reality device, and a robot.

24. The device of claim 1, wherein the at least one of the one or more AR content objects is determined based on movement of the device relative to the area of interest or the identified tile subarea.

25. The device of claim 24, wherein the at least one of the one or more AR content objects is auto-populated in the non-transitory computer-readable memory and rendered based on the movement of the device relative to the area of interest or the identified tile subarea.

26. The device of claim 1, wherein the identified tile subarea or the at least one of the one or more AR content objects is identified based on time.

27. The device of claim 1, wherein a modification of the at least one of the one or more AR content objects by a first user is viewable by a subset of users.

28. The device of claim 1, wherein a virtual item is made available to a user when the user executes a navigation event with respect to the area of interest or the identified tile subarea.

29. The device of claim 28, wherein the navigation event comprises at least one of the following: the user being located in the area of interest at any time, the user being located in the area of interest for a predetermined minimum amount of time, the user being located in the area of interest at a predetermined time, or the user being located in the area of interest and capturing an image viewable from the area of interest.

30. The device of claim 1, wherein a virtual item is provided to a first user and a second user when the first user and the second user execute a navigation event with respect to the area of interest or the identified tile subarea.

31. The device of claim 30, wherein the navigation event comprises the first user and the second user each performing at least one of the following: being located in the area of interest at any time, being located in the area

of interest for a predetermined minimum amount of time, being located in the area of interest at a predetermined time, or being located in the area of interest and capturing an image viewable from the area of interest.

32. The device of claim 1, wherein the at least one processor, upon execution of the software instructions, is further configurable to obtain the area tile map of the area of interest.

33. The device of claim 1, wherein the at least one processor, upon execution of the software instructions, is further configurable to generate the area tile map of the area of interest.

34. A method of presenting augmented reality (AR) with a computing device, the method comprising: obtaining sensor data from at least one sensor, the at least one sensor including a location sensor, wherein the sensor data includes a device location of the computing device obtained from the location sensor; obtaining an area of interest via an area database based on at least the device location within the sensor data; accessing an area tile map of the area of interest, the area tile map represented by a set of tile subareas that includes one or more tessellated tiles from a tessellated tile map; identifying a tile subarea from the set of tile subareas based at least in part on the device location relative to one or more locations of tile subareas from the set of tile subareas, wherein the identified tile subarea covers at least a portion of the area of interest, and wherein one or more tessellated tiles within the identified tile subarea are associated with one or more AR content objects; populating a non-transitory computer-readable memory of the computing device with at least one of the one or more AR content objects associated with the one or more tessellated tiles within the identified tile subarea to thereby accommodate capabilities of the non-transitory computer-readable memory via more accurately instantiating the at least one of the one or more AR content objects in the non-transitory computer-readable memory when the computing device is at or near the one or more tessellated tiles within the identified tile subarea; and rendering the at least one of the one or more AR content objects that is associated with the identified tile subarea on a display of the computing device based on a view of interest.

35. The method of claim 34, wherein the location sensor comprises at least one of the following location sensors: a global positioning system (GPS) sensor, an image sensor, and a wireless sensor.

36. The method of claim 34, wherein the device location includes at least one of the following: a global positioning system (GPS) location, a wireless signal location, a depth of field location, and an image-based simultaneous location and mapping (SLAM) location.

37. A computer program product embedded in at least one non-transitory computer-readable medium comprising software instructions for rendering augmented reality (AR) content on a computing device, which, when executed, configure one or more computer processors to perform a method comprising: obtaining sensor data from at least one sensor, the at least one sensor including a location sensor, wherein the sensor data includes a device location of the computing device obtained from the location sensor; obtaining an area of interest via an area database based on at least the device location within the sensor data; accessing an area tile map of the area of interest, the area tile map represented by a set of tile subareas that includes one or more tessellated tiles from a tessellated tile map; identifying a tile subarea from the set of tile subareas based at least in part on the device location relative to one or more locations of tile subareas from the set of tile subareas, wherein the identified tile subarea covers at least a portion of the area of interest, and wherein one or more tessellated tiles within the identified tile subarea are associated with one or more AR content objects; populating a non-transitory computer-readable memory of the computing device with at least one of the one or more AR content objects associated with the one or more tessellated tiles within the identified tile subarea to thereby accommodate capabilities of the non-transitory computer-readable memory via more accurately instantiating the at least one of the one or more AR content objects in the non-transitory computer-readable memory when the computing device is at or near the one or more tessellated tiles within the identified tile subarea; and rendering the at least one of the one or more AR content objects that is associated with the identified tile subarea on a display of the computing device based on a view of interest.

38. The computer program product of claim 37, wherein the location sensor comprises at least one of the following location sensors: a global positioning system (GPS) sensor, an image sensor, and a wireless sensor.

39. The computer program product of claim 37, wherein the device location includes at least one of the following: a global positioning system (GPS) location, a wireless signal location, a depth of field location, and an image-based simultaneous location and mapping (SLAM) location.
